



Machine Learning based intrusion detection systems for connected autonomous vehicles: A survey

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Abstract

Connected and Autonomous Vehicles (CAVs) expect to dramatically improve road safety and efficiency of the transportation system. However, CAVs can be vulnerable to attacks at different levels, e.g., attacks on intra-vehicle networks and inter-vehicle networks. Those malicious attacks not only result in loss of confidentiality and user privacy but also lead to more serious consequences such as bodily injury and loss of life. An intrusion detection system (IDS) is one of the most effective ways to monitor the operations of vehicles and networks, detect different types of attacks, and provide essential information to mitigate and remedy the effects of attacks. To ensure the safety of CAVs, it is extremely important to detect various attacks accurately in a timely fashion. The purpose of this survey is to provide a comprehensive review of available machine learning (ML) based IDS for intra-vehicle and inter-vehicle networks. Additionally, this paper discusses publicly available datasets for CAV and offers a summary of the many current testbeds and future research trends for connected vehicle environments.

Keywords CAN Bus · Intra-vehicle network · Inter-vehicle network · Intrusion Detection System (IDS) · Connected Autonomous Vehicles (CAV) · Security threats · Machine learning

1 Introduction

Connected and Autonomous Vehicles (CAVs) incorporating a variety of sensors, wireless communications, and onboard computing, enable vehicles to perceive the environment,

communicate with their surroundings, and control some or full driving functions [1, 2]. Each sensor in a CAV is linked to an Electronic Control Unit (ECU), which serves as a computational unit for connecting with the vehicle's On-Board Diagnostic Unit (OBD Unit). A modern high-end car has over 100 ECUs with computational power and is continuously connected to the Internet and other vehicles or servers through different interfaces such as Cellular Vehicle-to-Everything (C-V2X), and Dedicated Short-Range Communications (DSRC). With those functionalities, CAVs can achieve a certain level of automation and gain essential information for decision makings and are envisaged to dramatically improve road safety and efficiency of the transportation system, reduce environmental pollution, and increase accessibility to people such as seniors and people with disabilities. According to Boston Consulting Group, by 2025 the share of connected vehicles on the road will be 53%, and the Experiences Per Mile (EPM) Advisory Council predicts that by 2030, 96% of the vehicles shipped globally will be built with connectivity and 79% of new vehicles shipped globally will be equipped with L2 autonomy (where the control of acceleration, steering, braking in specific conditions are taken over by the vehicle) or greater [3].

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Such connected services and mobile applications mean increased attack surfaces, paving way for attackers to gain access to the network either passively or actively [4–7]. Like any network, attacks on connected car networks can be classified as passive and active. Passive attacks occur in the background, such as eavesdropping or data collection through a communication connection tapping. In an active attack, the adversary actually affects the operations in the attacked network. Among the active threats in a connected car environment are denial of service (DoS), false data injection, spoofing, Sybil attacks, and so on. For instance, in 2015, Miller and Valasek [8] demonstrated how an unprepared cybersecurity system might jeopardize driver safety by remotely controlling the audio, windshield wipers, steering, and brake systems of a moving Jeep Cherokee. Additionally, Keen Security Lab hacked Tesla and BMW vehicles [9] to showcase security concerns and prospective attacks against connected automobiles. Thus, having a well-defined and consistent architecture - both within the vehicle and in the cloud - is crucial for Original Equipment Manufacturers (OEMs) and other mobility sector participants to ensure the security of these increasingly complex ecosystems [10, 11].

Communication in a connected vehicle can occur over two segments: intra-vehicle or in-vehicle network, and inter-vehicle network [12, 13]. The in-vehicle communication network is made up of many sensors communicating with one another through Controller Area Network (CAN) bus, Flex ray, Ethernet, Media Oriented Systems Transport (MOST) and Local Interconnect Network (LIN). Because of their construction, which was meant to decrease cabling in the car, these communication routes are sensitive to assaults. Since its introduction, the CAN bus has been the preferred communication protocol due to its dependability, low cost, and ease of communication with minimal wiring. Inter-vehicle techniques link the vehicle to the external environment, which includes other vehicles, pedestrians, infrastructure, roadside units, and more, and are referred to as V2X. VANET (Vehicular Ad hoc NETWORK) is a component of a larger ecosystem that includes vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-pedestrian (V2P), and vehicle-to-everything (V2E) communication [14, 15]. The CAV's goal is to give its passengers a secure, organized, time-saving, information-rich, digital experience using all of the data acquired from its surroundings. The main problem, however, is ensuring the security and dependability of all information transferred inside its communication network. A single occurrence can change the fate of passengers as well as those in the surroundings. As an outcome, the attack is mitigated, and similar attacks in the future are identified and prevented.

These security threats in CAV networks are addressed through (i) Prevention: A proposed list of defense strategies that identify the threat and prevents it from happening; (ii)

Detection: If the system is unable to avoid the danger during the prevention stage, the threat detection model should be able to quickly identify the attack, the nodes involved in the attack; and (iii) Mitigation: Responding to an attack, countermeasures are used to overcome impacted nodes, and secure the vehicle.

Intrusion Prevention Systems (IPS) aid in the prevention of targeted attacks; nonetheless, an attacker may be able to access the network through this system. IPS can prevent previously modeled assaults, but it is unable to recognize unique attacks, hence an intrusion detection system is used as a second line of protection. An IDS in a security strategy identifies the attacker, the attacker's position, the time of the attack, the location, the type of intrusion activity, the layer in which the assault occurred, and the nodes affected by attack. Thus IDS can be defined as a hardware and/or software system designed to identify any combination of activities intended to threaten a resource's confidentiality, integrity, or availability. This technique can be categorized based on the strategy implemented. Anomaly detection is based on statistical behavior modeling, in which the intrusions are known members, and any divergence from typical behavior are marked as an anomaly. The drawback of this method is that it may result in more false positives, causing havoc in the system. False Positive defines an activity as an attack, although it is a permitted behavior within the system, whereas True Positive identifies an activity as an attack and this behavior is actually an attack. Attacks that have already had a signature established can be detected by systems that are signature-based. This method has the disadvantage of not identifying any new attacks that were not registered and assigned a signature beforehand. IDS in automotive domain focuses on detection at central gateways, ECUs and CAN bus and OBD unit. With the implementation of ISO 26262 - Functional Safety Standards and integration of ISO/SAE 21434 (evolution of SAE J3061)- Road Vehicles Cybersecurity Engineering, ECUs are rendered with additional cybersecurity components making them host based. The benefits of ML-based approaches include their capacity to construct explicit and implicit models of examined patterns, which are updated on a regular basis to increase detection performance based on prior findings utilizing Bayesian networks, Markov models, Fuzzy logic, and other techniques.

This survey paper on machine learning-based intrusion detection in connected autonomous vehicles stands out from other surveys in the field due to its specific focus on ML-based approaches and its coverage of the unique challenges and considerations relevant to connected autonomous vehicles. While other surveys, including but not limited to [5, 16, 17], may touch on the topic of intrusion detection for connected vehicles, they often do not delve into the ML-specific techniques and algorithms that are increasingly being utilized in this area. Additionally, many existing surveys focus

on either in-vehicle networks (e.g. CAN bus) [18–20] or VANETs [21], whereas this survey paper takes a comprehensive and systematic approach that considers IDS solutions designed for both types of networks. Moreover, this survey provides an in-depth review of datasets and simulation environments for in-vehicle and inter-vehicle domains, along with available solutions for creating physical, virtual, and hybrid testbeds. Overall, the ML-centric and holistic approach of this survey paper makes it a valuable resource for researchers and practitioners working in the field of connected autonomous vehicle security and assists them in making informed decisions regarding the development and evaluation of autonomous driving systems.

The aim of this paper is to present a comprehensive review of A) Intrusion Detection Systems based on machine learning for in-vehicle and inter-vehicle communications; B) Various publicly available datasets for IDS in CAVs; and C) Configurations of test beds for vehicular networks that are currently available on the market. As shown in Fig. 1, the structure of the paper is as follows. Section 2 provides a high-level overview of the in-vehicle network, including a description of the serial data exchange protocol, in-vehicle threats, and an analysis of machine learning-based intrusion detection systems. A section on inter-vehicle networks is presented in Sect. 3, while Sect. 4 discusses datasets and testbeds, with future trends and a summary in Sects. 5 and 6.

2 In-vehicle network: security threats and IDS

2.1 In-vehicle network

Knowledge of serial bus protocols enables a thorough understanding of in-vehicle networks and the modeling of associated threat prevention mechanisms. A modern vehicle is comprised of several serial bus protocols: the CAN bus, the Local Interconnect Network (LIN), the FlexRay, the Media Oriented System Transport (MOST), and Ethernet. Serial bus protocol manages the powertrain, chassis, body, telematics, and infotainment domains by establishing a network between various gateways [22].

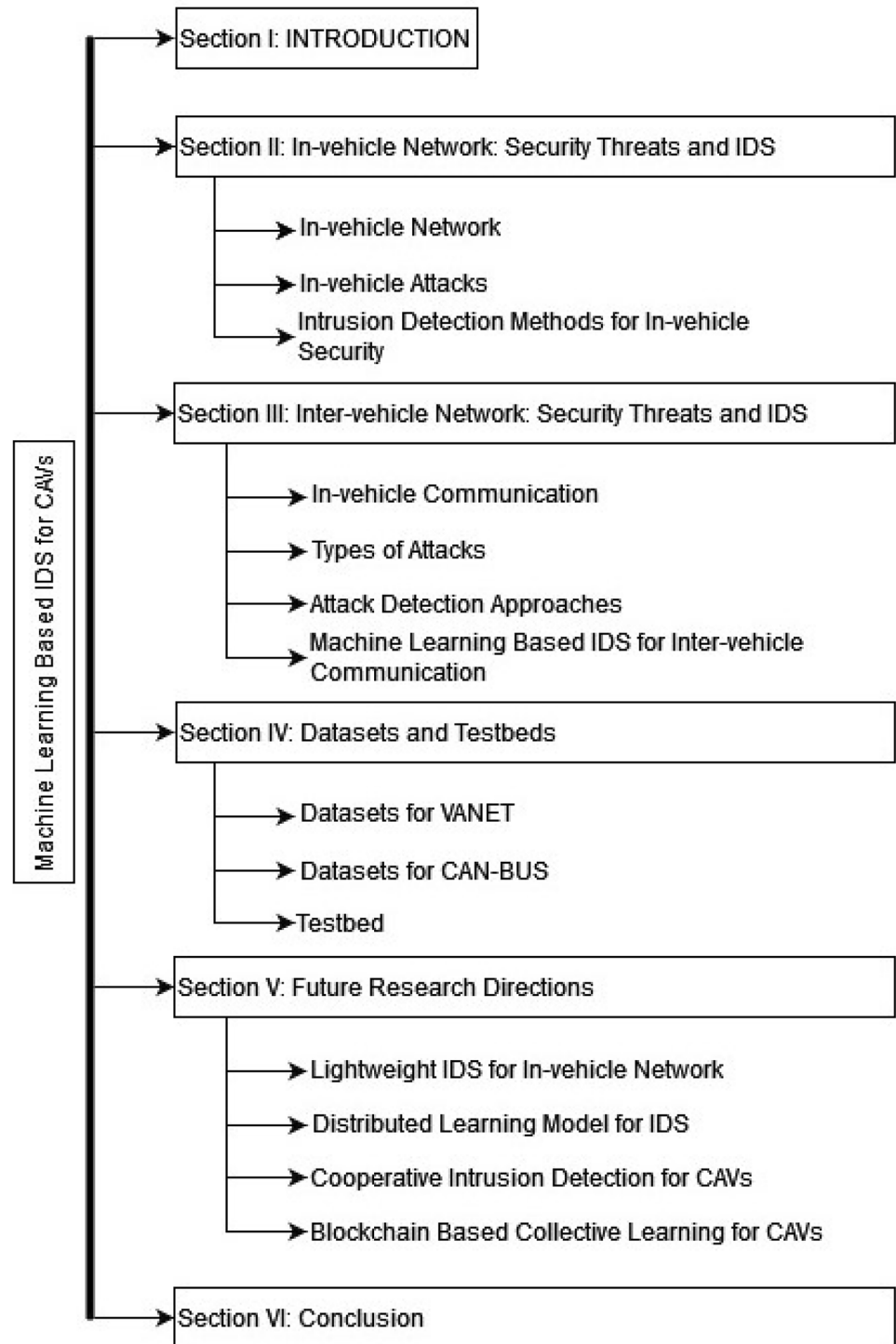
The Local Interconnect Network (LIN) protocol is a serial transmission protocol, used in low-cost, low-bitrate, and non-critical applications such as door modules, tailgate modules, roofs, air conditioning, and lumbar support systems. A master–slave architecture is used in the LIN protocol to cluster the nodes in a bus topology. The master node controls when and which slave nodes are allowed to access and transfer their data frame on the bus. The message frame consists of a header, provided by the master, and a response, which is provided by either the master or the slave. It can accommodate up to 16 nodes and works at a maximum bit rate of 20 kbps [23].

FlexRay protocol is an automotive communication protocol that enables time-synchronized data exchange between ECUs and prevents data corruption on a given network within the communication system. There are three standard bit rates supported by FlexRay - 10, 5, and 2.5 Mbps. The protocol supports both star and bus topologies. As part of FlexRay's fault tolerance feature, nodes are connected to two redundant communication channels, one passive and one active. It is possible to allocate channels to the same or different topologies, thereby offering a robust and flexible hybrid network. Its robustness and secure message transfer enable it to be used in mission-critical and safety-critical systems such as adaptive cruise control, high-performance engine control systems, active suspension systems, and throttle exchange systems, all of which require a high volume of data exchange [24].

The MOST protocol serves as the foundation for current and future automotive multimedia networks that transmit multimedia and control data through the use of fiber optic cables. A maximum of 150 Mbps can be achieved with this plug-and-play network that is capable of connecting up to 64 vehicles in a ring, star, or bus topology. MOST enables synchronous and asynchronous data transfer, as well as a limited control channel, similar to FlexRay and the messages include the address of the sender and receiver. Error management is handled internally by a MOST system service, which detects errors in parity bits, status flags, and checksums and, if necessary, disconnects erroneous nodes. MOST is vulnerable to synchronization disruption and jamming attacks [25].

CAN is an asynchronous, multi-master communication protocol that is used to connect between 50 and 100 Electronic Control Units (ECUs) in a modern automobile. The CAN bus protocol enables reliable communication, reduced cabling, and reduced overall system cost through a distributed network architecture that provides uncomplicated support. It supports a range of high-speed network applications with up to 1 Mbps data rate to low-speed and low-cost networks. Depending on the required specifications of applications, different CAN buses might exist in a vehicle and connect through gateways. Due to its simplicity and reliability, CAN is the dominant in-vehicle communication protocol [26].

Messages on the CAN bus are transmitted through the CAN data frame. A frame consists of seven fields: the Start of Frame (SOF), an arbitration field, a control field, a data field, a CRC field, an ACK field, and an End of Frame (EOF). Due to CAN's multi-master nature, all nodes connected to the bus listen to it and transmit their frames once it is idle. When a node accesses the medium and starts transmission, others sense the SOF, a single dominant bit, and back off. The arbitration field, or the identifier, is an 11-bit field, determining not the receiver but the meaning of the

Fig. 1 Paper organization

frame. The control field consists of two reserved bits for future use followed by a 4-bit Data Length Control (DLC), which indicates the length of the data field. The Data field carries the node's generated data and can be up to 8 bytes. By acknowledging a valid frame, the CRC and ACK fields contribute to reliable frame transmission.

The primary advantage of the CAN protocol is the differential wiring, which provides increased immunity to noise and electrical interface. The CAN architecture was designed to be light and resilient, without segments, encryption, or authentication, allowing CAN messages to flow freely between and among ECUs. These characteristics, however,

directly contribute to CAN's security flaws. Each ECU broadcasts and receives all data transmitted over the CAN bus and then determines which messages are addressed to it. CAN is designed in such a way that it cannot prevent unauthorized devices from connecting to the bus and transmitting harmful messages to all ECUs.

Besides the vehicle-specific protocols, Ethernet, a high-speed, general-purpose, flexible, and scalable communication protocol, is used for the in-vehicle network. Ethernet has been the main transportation protocol for Local Area Networks (LAN) since its invention and is already used in V2X applications such as EV charging control modules, diagnostics, and ECU software updates. By reusing Ethernet in in-vehicle domains as well as inter-vehicle, the current heterogeneous automotive network, consisting of various networking technologies, will emerge and become more manageable and scalable. It provides a high bandwidth of up to 100 Mbps, which is ten times more than FlexRay, and allows a hierarchical architecture for connecting ECUs. Due to its best effort and random media access nature, Ethernet cannot guarantee latency or minimum bandwidth, making it less suitable for safety-related and delay-sensitive applications, such as Autonomous Driving Assistant Systems (ADAS). Some extensions to Ethernet, like Automotive Ethernet (AE) [27], Audio/Video Bridging (AVB) [28, 29] and Time-Triggered Ethernet (TTEthernet) [30] are proposed to provide Quality-of-Service (QoS) for such applications.

Among the mentioned In-vehicle communication protocols, CAN is the dominant one, and its security is challenging. In the following sections, we will discuss the security challenges and threats on the CAN bus, and then a comprehensive overview of the related works on developing attack detection strategies will be given.

2.2 In-vehicle attacks

An in-vehicle network connects numerous electrical components such as ultrasonic radar, lidar, camera, and other sensors. Each sensor type has its own range, detection capability, and dependability. External entities can also connect to sensors using existing wireless access methods. Due to connectivity, an attacker can access a susceptible and peripheral sensor of a CAV, and then control the sensor and launch various remote attacks [31].

2.2.1 CAN bus spoofing attack

As CAN protocol does not provide authentication and encryption procedures, it is feasible to tap a compromised node into the CAN bus and transmit forged messages with an existing identifier in the network. By doing so, an attacker disrupts a function in the vehicle by generating invalid

messages targeted at it. The spoofing attack can be prevented by implementing node authentication and message encryption; however, it puts a computational burden on ECUs.

2.2.2 CAN bus fuzzing attack

ECUs rely on CAN bus communications to function effectively, but since the CAN bus messages lack authentication, the ECUs cannot verify the validity of the message's source. As a result, when a random CAN bus frame is introduced aimed at a specific area in the car, the ECU finds it challenging to distinguish between authentic and fraudulent messages, resulting in a fuzzing attack. The aim of an attacker in the fuzzing attack is to examine what happens in the ECUs while also tracking the attack's purpose. For example, if the attacker seeks to change the vehicle's speed, it monitors the communication between the targeted ECU and the CAN bus as well as the change in vehicle speed when the assault is carried out. Additionally, the CAN bus can be regarded as a black box by the attacker, in which the CAN ID and payload values are generated randomly, with no prior knowledge of the actual CAN IDs in the system. It consists of transmitting randomly generated CAN frames to disrupt the vehicle's operations. To prevent data analysis from occurring, encryption and authentication are required, with the latter ensuring that only valid ECUs transmit CAN frames in the first place.

2.2.3 CAN bus frame falsifying attack

Frame falsification attacks occur when wrong values are added to messages in the CAN Bus payload. As an illustration, an attacker may introduce erroneous parameter values into a vehicle. This attack is once again a result of the lack of authentication and encryption in a CAN bus architecture. Authentication mechanisms allow users to authenticate the source of data before acting on it to counteract this type of assault. In order to identify this attack, a system should consider evaluating the consistency of the CAN ID and the data payload over a while.

2.2.4 CAN bus injection attack

Without encryption and authentication, any node can connect to the bus and monitor the bus for transferred arbitration and data fields. Also, once a malicious node gains access to the CAN bus, it is possible for it to transmit targeted or massive data, allowing modifications to the frequency and altering the CAN bus frames. This could cause fraudulent events to be generated, causing portions of the vehicle to act as desired. Authentication and integrity protection methods can stop this attack.

2.2.5 CAN bus DoS attack

ECUs connected to the CAN bus compete to access the channel based on CAN ID priority. The arbitration field is used by the CAN Bus nodes to determine message priority and which node is allowed to use the bus to convey information. Thus, using the highest possible attribution id, such as 0x000, an attack can be launched to occupy the bus and prevent other nodes from using it. Knowing the transmission rate of an existing ECU's CAN message ID, one can launch a DoS attack by increasing the frequency of injected messages. Consequently, if a legitimate ECU sends messages every 200 milliseconds, an attacker can insert messages with more dominant bits in the ID field at a greater frequency, producing sensor disturbance and generating a successful DoS attack.

2.3 Intrusion detection methods for in-vehicle security

In the early stages of introducing anomaly detection methods into the in-vehicle CAN network, most of the research focused on the statistical modeling of the CAN bus behavior based on the deterministic frequency of CAN IDs in the messages. ECUs connected to the CAN bus tend to generate messages periodically at a specified time, broadcasting their messages containing one or more sensor statuses in the vehicle. In 2015, Taylor et al. proposed a frequency-based anomaly detection, considering the average and standard deviation of inter-packet timings in normal traffic flow and its deviation in the under-attack situation. They also conducted the evaluation of their method using the Hamming distance between successive message data fields and its combination with timing difference, but they found them unreliable in detecting anomalies in sequences. They evaluated the efficiency of their method on the data collected from a 2011 Ford Explorer and compared it with the OCSVM classifier while simulating Replay and Message Erasion attacks. They could achieve more than 98% AUC when the rate and duration of attack increased to 1x and 0.5 s, but their method fails when it happens in a smaller period. They, however, found that OCSVM works better in these cases [32].

Marchetti and Stabili proposed another heuristic approach in 2017 to detect CAN anomalies by focusing on possible CAN ID sequences in normal data. They introduced a so-called Transition Matrix, initiated to all False values, to keep the recording of all the legitimate consecutive IDs marked as True elements in it. Their method was validated against two generated datasets, containing Replay, Bad Injection (injecting never-seen ID sequences), and Mixed Injection (injecting a random mix of ID sequences) attacks. It performed very well on Bad and Mixed Injections but failed on Replay attacks. The reason lies in the fact that the replayed CAN messages

were already seen in the normal data, making them unrecognizable in the detection phase [33].

OTIDS, presented by Lee et al., is also an IDS designed based on timing specifications between the remote frame and its response. An entity on the CAN bus can request an ECU to send a requested signal encoded in a CAN message by sending a remote frame. In this IDS, a detection module is connected to the bus and sends remote frames targeted to all ECUs. The time interval and offset (number of transmitted messages between the remote frame and associated data frame) are recorded, as well as their correlation coefficient, and used as the feature for detecting anomalies in a publicly available dataset, named CAN Dataset for intrusion detection. The idea behind the proposed method is that when an attacker injects messages into the bus, it could cause delaying response data frames and be detected by analyzing it. They claim that the OTIDS can detect DoS, Fuzzy, and Impersonation attacks, although they didn't provide a performance metric to support it. Another drawback of this method is that it burdens the CAN bus with extra remote frames and their responses, which may degrade its throughput for legitimate messages [34].

In the literature on CAN bus anomaly detection, data mining is another technique for extracting meaningful patterns from raw data. Rieke et al. presented an anomalous behavior detection system based on data mining and Petri Nets. The valid sequence of events is extracted in the discovery phase using a process mining technique - called the Alpha algorithm. In the conformance checking phase, if the given transition between events does not comply with the derived behavioral model, it is marked as an anomaly [35].

Statistical approaches were the main methodology for anomaly detection by 2018. Even so, it has maintained its popularity among researchers in recent years. Mo et al. adopted statistical thinking to the problem and measured the similarity of the decimal-converted data field of CAN messages with the same IDs, using the Mahalanobis distance. Then a confidence interval is determined based on the average and standard deviation of distances in the normal data population and used as a detector in data containing attacks, achieving more than 90% detection rate [36]. In another study conducted by Sunny et al., a hybrid approach is presented to take the CAN ID sequence and the timing of messages into account for detecting abnormal situations. A so-called sequence tree is extracted in this method which represents the valid CAN ID sequences in a sliding window, as well as the maximum and minimum authorized time interval between two consecutive messages with the same ID. In the test procedure, if the observed ID trace does not exist in the sequence tree or the occurrence timing does not satisfy the time interval condition, it is flagged as anomaly [37].

Although the heuristic and statistic-based anomaly detection methods seem to be lightweight and resource-friendly,

especially concerning the real-time detection requirement in the CAN bus communications, they lack the ability to consider anomalies that occurred in the message data payloads. Almost all the mentioned works focus on the deviation in CAN ID and time spacing of messages. As opposed to the message injection types of attacks, such as DoS and Fuzzy attacks, that change the frequency of messages, a victim function of the vehicle can be targeted and interrupted by manipulating the content of messages, causing vital damage to the vehicle, driver, and others. These kinds of malicious activities only can be detected by applying machine learning to the detection system. Newer research programs, mainly after 2018, have integrated machine learning methods into their work and achieved higher performance.

Traditional ML algorithms and neural networks are the two main design categories in machine learning-based anomaly detection. While traditional ML methods are rather faster and simpler in structure and have fewer hyperparameters to tune, neural networks, on the other hand, simulate how the human brain learns from the unstructured data and makes better decisions with less intervention in the learning process. There are works done in both categories, with a trend in applying neural networks in the most recent ones, which we will discuss in detail.

2.3.1 Traditional machine learning methods

OCSVM, a one-class extension of the SVM classifier, is one of the state-of-the-art traditional novelty or anomaly detection methods. In work by Theissler, a modified OCSVM is used to classify time-sequenced data into normal and anomaly classes. Due to its one-class nature, it is trained with the attack-free or majority class of data to find the tightest hyperplane to separate them from the outside world. To make it applicable for CAN messages with a sequence in time, the author has defined a non-overlapping window over the time series and modified the distance metric of the OCSVM to take an average between the distance metric of instances in the window. The effectiveness of this method in detecting conditional anomalies in the sequential data was proved by testing collected data from various driving conditions. The reported results indicate 76% precision and 59% TPR (True Positive Rate); however, the anomaly situations are not described clearly in this work [38].

The detection performance of OCSVM merely depends on properly adjusting its hyperparameters, namely the kernel type, function, and parameters. To select the best hyperparameter values in the training phase of OCSVM, an optimizer with a smart exploration space and algorithm can be used. Therefore, Avatefipour et al. proposed a Modified Bat Algorithm (MBA) to optimize the parameter values to achieve optimal performance [39]. In addition, Al-Saud et al. integrated another optimization algorithm based on the

modified Social Spider Optimization (SSO) into OCSVM [40]. Both works aim to alert the DoS attack that happened in a real car by applying their method to the CAN ID and frequency of messages. The results showed that the proposed anomaly detectors by the authors have higher performance for detecting anomalies in the CAN bus compared to the classical OCSVM, the Isolation Forest, and other baselines in terms of Receiver Operating Characteristic (ROC) curve, hit rate, miss rate, false alarm rate, and correct reject rate metrics. There is, however, no testing of the proposed method against other types of attacks, and it might be insufficient when the data field content is changed rather than the message frequency of the victim ECU.

HMMs (Hidden Markov Models) are powerful traditional machine learning tools that can be used to analyze the temporal sequence of events. Unlike other applications, transmitted CAN messages are not individually independent but rely on the previous flow of observed messages on the CAN bus. HMM is a statistical model also used in machine learning to extract the sequential behavior of the system by defining states and transitions and calculating the probability of occurring an event based on them. It is utilized in the structure of OBD-SecureAlert, an anomaly detection system proposed by Narayanan et al. They generated the HMM model with the Baum-Welch algorithm and tested it against sudden anomalous changes that happened in RPM and speed data. An alarm is issued when the posterior probability of observing a sequence falls below a defined threshold, meaning it is not likely to happen [41].

Another HMM-based novelty detection approach is designed by Levi et al. Instead of raw CAN messages, the authors added an abstraction layer to the data by integrating CAN, OS, and network interfaces and defined events, such as opening the door and starting the engine, and stories - a chain of events. They also redefined attack scenarios and included out-of-order, out-of-context, USB firmware update attacks, communication with unknown vendors, Over The Air (OTA) malicious updates, and malicious application installation. The HMM model is trained on the normal abstract data, outputting the log-likelihood probability of sequences. Then, a linear regression model is trained to determine the detection probability threshold [42].

Casillo et al. suggested an embedded system intrusion detection system for the automobile industry that takes advantage of the ability of Bayesian networks to detect cyber attacks on the vehicle's CAN-Based communication system. The entire experimental phase was carried out with the help of an automobile simulator, CARLA, which allows for the simultaneous execution of hardware and software testing in order to ensure that new automotive components operate correctly. They simulated three attack scenarios by manipulating the content of throttle, brake, speed, and steer and achieved about 80% precision, recall, and F-score [43].

Tian et al. suggested the Decision Tree (DT) to solve the problem and Gradient Boosting (GB) technique to increase the detection rate. Based on the CAN ID and frequency of messages, they constructed a new entropy-based feature and added it to 8 bytes of data fields. The method could achieve 97.67% TP and 1.2% FP by separating the data collected from an actual car according to the CAN ID field, applying DT to each of them, and combining them using GB to get a result on the aberration of the input data. But, the results are not compared to other methods to prove its superiority [44].

Statistics and traditional machine learning tools are combined in the event-triggered anomaly detection method offered by Han et al. The authors measured the interval between two following messages of the same CAN ID by the number of transferred messages, called events, instead of elapsed time. First, the anomaly detection module discovers whether a message is abnormal by comparing it to an extracted event-triggered interval threshold for each CAN ID. Then five statistical features, including mean, variance, Q1, Q3, IQR, skewness, and kurtosis, are measured for intervals and fed into a multi-class classifier to recognize the type of attack. The proposed method is validated against two datasets gathered from real cars and with three classifiers, namely decision tree (DT), RandomForest, and XGBoost [45].

A combinational hybrid IDS method, called MTH-IDS, is also proposed by Yang et al. It consists of signature-based and anomaly-detection parts, detecting attacks in both in-vehicle and inter-vehicle domains. The signature-based module is a DT-based machine learning model, trained with known and labeled attack samples. On the other hand, a k-means clustering algorithm is employed in the anomaly detection module to group similar data samples in an unsupervised manner, representing the normal situation, and declare any data falling outside the cluster as an anomaly. Several optimizations and feature engineering approaches were applied to MTH-IDS to improve its detection performance [46].

2.3.2 Neural networks

By mimicking how the human brain neurons learn from observations and make a decision based on them, Neural Networks (NN) paved their way into the decision-making problems traditionally done by humans, including in-vehicle attack detection. Neural networks are a network of connected nodes in a layered structure. Each layer processes its inputs using mathematical operations and passes its results to the next layer. There are three types of NNs, namely feed-forward, recurrent, and convolutional. Each of them is best suited to a particular kind of application.

Feed-forward neural networks Kang and Kang were among the first researchers who introduced neural networks into CAN bus anomaly detection in 2016. The proposed method

consists of a feed-forward deep neural network with several hidden layers. The hidden layers are pre-trained with the Restricted Boltzmann Machine to an initial value in order to prevent the gradient vanishing problem. The rest of the training is done with a feature set of bit-symbol occurrences representing signals and evaluated against spoofed Tire Pressure Monitoring System (TPMS) data. The detection rate is reported at 99.9% on a seven hidden layers deep network. Thus, through their model, the authors proved that deep learning-based IDS can be more robust and effective for identifying the most difficult attacks [47].

Zhang et al. suggested the Gradient Descent with Momentum and adaptive gain (GDM/AG), an enhancement of the Back Propagation (BP) algorithm, as an optimizer for the training process of DNN to increase the efficiency and accuracy of their DNN-based IDS. Tests of the model's TPR, FPR, and process time were conducted on actual automobiles. Experiments demonstrate that the GDM/AG algorithm helps the DNN to converge faster than the GDM algorithm and detect in-vehicle anomalies in milliseconds. Their model can also adapt to unknown attacks. When facing spoofing and replay attack types, the true positive rate is 97–98 percent, and the false positive rate is under two percent [48].

Hierarchical Temporal Memory (HTM) is a machine learning algorithm developed by Numenta®, based on the neocortex interactions of the brain. In its tree-shaped structure, HTM models the sequential flow of data using memory. An intrusion detection system relying on HTM was offered by Wang et al. The proposed method adopts an online learning scheme. 64-bit stream of CAN message data fields from the Impreza high-speed bus was logged and continuously fed into the model to train and predict the next expected value of data payloads. For each of the CAN ID streams, an anomaly score is calculated and aggregated among different streams in order to get a consensus view on the status of the current data. The authors contrasted the HTM detection performance with RNN and HMM using precision, recall, and AUC metrics. It outperformed the competitors when facing the Replay and Spoofing attacks [49].

To identify attack messages sent on a CAN bus, Barletta et al. proposed a high-performance IDS based on an unsupervised Kohonen Self-Organizing Map (SOM). SOM is a feedforward neural network, introduced by Teuvo Kohonen, trained not with error correction but the competitive learning [50]. It is commonly used for dimension reduction and clustering of data in an unsupervised manner. The authors chose this method and combined it with the k-means clustering algorithm because it has a high detection rate, requires less time to train, and can be used in a variety of applications. The model was evaluated on the Car Hacking Dataset, an open source dataset of CAN bus 2.0B data packets, with high traffic volume and four attack types, including DoS, Fuzzy, RPM Spoofing, and Gear Spoofing attacks [51]. The

proposed architecture had high accuracy in detecting DoS and Spoofing attack messages but failed in detecting Fuzzy attacks [52].

Convolutional Neural Networks In addition to the feed-forward NN, the other type of neural network studied in the CAN bus IDS literature is the Convolutional Neural Network, CNN for short. CNN benefits the spatial correlation in the two-dimension data, for example, image frames, by moving a filter on it, left to right and top to bottom, and applying convolution operation between the filter and the overlapping area, known as the receptive field. Song et al. developed a deep CNN, called Inception ResNet, and extracted the spatial bit-wise sequence of CAN IDs in normal and under-attack situations. By doing so, they could detect DoS, Fuzzy, RPM Spoofing, and Gear Spoofing attacks almost perfectly. On the downside, due to the complex and deep design of the model, the convergence and detection time is higher than other available baselines, such as LSTM [51].

Mehedi et al. implemented a CNN attack classifier, LeNet, for attack detection in IVN. They chose CAN ID, DLC, and Data fields in the CAN message as the feature set and fed it into two 1D CNN layers, two pooling layers, a flattened layer, and a dense layer which outputs the class of data: normal, flooding attack, spoofing attack, or fuzzing attack. The performance of LeNet was compared with that of traditional machine learning and deep learning algorithms using the Car Hacking dataset [53].

Recurrent Neural Networks Recurrent Neural Network (RNN) is another category of NN, useful for predicting time series. Due to the sequential nature of data generated by each CAN ID, RNN is the widely used tool in the CAN bus anomaly detection area. Many authors selected RNN as their primary methodology and applied it to various features and datasets, with distinct architectures and combinations with other techniques. Taylor et al. were the first researchers to adopt Long Short Term Memory (LSTM), a type of RNN, to learn the long-term dependencies within the ECU data. The architecture includes two linear layers, followed by two LSTM layers, to predict the current binary data field value transmitted by each CAN ID. The binary log loss error is measured between the predicted value and the observed one and merged by maximum bit loss as the reported anomaly score. They determined a threshold for anomaly scores by minimizing false alarms on the attack-free data and evaluated the True Positive Rate (TPR) in data containing discontinuity, drop, interleave, reverse, and unusual types of attacks. Although it was the first LSTM-based detection system in the literature, the proposed method did not work well for attacks on some CAN IDs, motivating other researchers to improve it [54].

To decrease the detection time and computational overhead of the LSTM-based IDS, Zhu et al. proposed a mobile

edge-assisted anomaly detection method. This work incorporates both time interval and data fields as features while it is distributed on edge devices close to vehicles. They utilized almost the same architecture offered in [54], with a difference in feature set, which is a combination of normalized time interval and the 64-bit binary data field. They used the cross entropy error, a generalized version of log loss error, to judge the anomaly of the observed data. In the distributed framework, the two first linear layers are trained locally, and the rest LSTM layers are done in the edge server. This would reduce the training power and time since LSTM training is running in parallel on different processors. The performance evaluation process showed 90% accuracy and 0.61 milliseconds detection time, which was better than the compared baselines [55].

Focusing on the computational power of processors in the CAN bus, Xiao et al. presented a new RNN unit, called the Security Control Unit (SECCU), by combining the power of LSTM into the customized GRU layout. The authors also adopted a simplified attention (SIMATT) model into the approach to make it lightweight and avoid the unnecessary complexity of the model. SECCU was compared with GRU and LSTM on a ROC curve, along with SIMATT's influence. The simplified and lightweight composition of SIMATT-SECCU outperformed other baselines and dropped the computational overhead [56].

Hossain et al. proposed two multi-class attack detection methods based on LSTM and CNN in [57] and [58], respectively. The LSTM-based method is made up of one LSTM layer followed by a dense layer with a softmax activation function to classify messages into four classes: Benign, DoS, Fuzzy, and Spoofed. Similarly, in the CNN-based method, two 1D CNN layers and a dense layer are applied to the time series data. To evaluate the CNN approach, three OEMs such as Toyota, Subaru, and Suzuki provided in-vehicle CAN bus data, whereas the LSTM classifier was tested against NAIST data and automobile IDS survival analysis data. The authors identified the optimal hyper-parameter settings based on both structures for enhancing the detection rate. According to the findings of the experiments, they were able to establish a high assault detection rate for the CAN bus network. Due to their nature as attack detectors compared to anomaly detectors, both cannot detect unknown attacks other than those mentioned above.

In [59], Balaji and Ghaderi suggested incorporating the inter-ECU dependencies into the prediction of the upcoming CAN message, as a means of anomaly detection, by adding an embedding layer. The embedding layer is a linear transformation of the input data from each CAN ID, passed through a sigmoid function and accumulated over all IDs. It is then followed by an LSTM and an output layer, shaping a prediction-based anomaly detector - called Neuro-CAN. With the embedding layer, LSTM showed improved

detection accuracy. The network is trained separately for each CAN ID, resulting in high memory and computational cost. Instead of doing so, later, the authors improved NeuroCAN and introduced CANLite, in which a single multi-task model is learned. This model extracts the spatial and temporal dependencies of CAN IDs and predicts the next payload of a requested CAN ID based on a prediction parameter. This makes CANLite a more lightweight contextual anomaly detection system, better suited for CAN buses with restricted resources and detection time [60].

Combinational Neural Networks CNN (for extracting spatial patterns) and RNN (for learning temporal correlations) are powerful neural network tools on their own, but when combined, they can boost the dimensionality of features and the performance of detection. The CANintelliIDS methodology proposed by Javed et al. employs a combination of CNN and Gated Recurrent Unit (GRU), an RNN unit. The CNN part of the network includes two convolution and two dense layers, whereas the RNN part is a three-layer deep Attention-based GRU (AGRU). The GRU is a newer class of RNN, very similar to LSTM, however, it has two gates - the update gate and the reset gate - making it more efficient than LSTM. Additionally, an attention mechanism has been introduced in RNN to help it act more like the human brain, focusing on what is more important while ignoring what is not. An attention-based GRU assigns a weight to hidden states based on their importance in the final prediction. The authors have tested the CANintelliIDS model extensively on various scenarios identifying DoS, Fuzzy, and impersonation attacks, and it worked better than conventional ML methods, DNN, and simple CNN, in terms of precision, recall, and F-score [61].

The CLAM model offered by Sun et al. is also a fusion of 1D CNN and attention-based bidirectional LSTM. It first extracts the multivariate signals from raw binary CAN messages and then exploits the local features in the CNN layer as well as the contextual features of the input signals with Bi-LSTM units. The outputs from the Bi-LSTM layer are weighted using the attention mechanism and pass through a dense layer to make prediction [62]. Thiruloga et al. claimed that a very long-term dependency exists in the messages transferred between ECUs that LSTM and GRU cannot capture. So they designed Temporal CNN with Attention (TCNA) for predicting such time series. Three TCNA blocks are stacked in the proposed structure, named TENET, each consisting of a TRB (TCN Residual Block) and a self-attention block. The TRB itself is a residual network of two DCC (Dilated Causal Convolution), two Weight Normalization, and two ReLU layers. This well-tuned structure decreased the number of trainable parameters, hence the training and detection time, while adopting longer dependencies in the CAN signals [63].

Autoencoder neural networks The largest group of neural network structures offered in the anomaly detection literature is the Autoencoders (AE). Autoencoder is a deep neural network that extracts a compressed representation of the input data in the encoder part and then reconstructs the input using the encoded model in the decoder part. Since CAN bus behavior is normally represented by the compressed model, it would be very difficult to reconstruct the anomalous data if it were fed this model.

Lokman et al. proposed a deep contractive neural network AE (DCAE) regularized with the Frobenius norm of the Jacobian matrix. The regularization term improves the robustness of the perceived model to small changes in the data and avoids misdetection of normal faulty electronic devices as anomalies caused by attacks [64]. To tackle the same problem, Lin et al. implemented a deep denoising autoencoder (DDAE). Some noise is added to the data before inputting it to AE to enhance the stability of prediction. As another contribution, this model integrates an evolutionary-based optimization algorithm known as Ecogeography Based Optimization (EBO) into its DNN autoencoder to provide optimum training structure as well as to avoid premature convergence. It was used to identify and detect DoS, Fuzzy, and impersonation attacks when tested on three different real-world data sets where this model outperformed other ML models by achieving the highest False Positive Rate and lowest True Negative Rate [65].

In a work by Jaoudi et al., a CNN-based AE is designed for anomaly detection in the CAN bus. Three layers of 1D convolution compress the data, and three transpose convolution layers decode it again. The authors converted the model into a Spiking Neural Network (SNN) using the NengoLoihi package and tested it on the Loihi chip. The proposed anomaly detector had higher precision, recall, and F-score compared to OCSVM and Isolation Forest [66].

Longari et al. proposed CANNolo, which uses LSTM autoencoders to detect abnormalities in the flow of the CAN bus network. This method consists of two main components, mainly an LSTM autoencoder, which is a trainable model which can master the behavior of normal signals, and the second part is an anomaly detector that helps to analyze the reconstruction errors. The disadvantage of this method is relatively slow computation [67].

By employing LSTM and Exponential Line Units (ELU) in an Autoencoder architecture, Hanselmann et al. suggested a novel deep learning-based IDS, called CANet. Unlike previous works, this technique could process multiple signals carried in messages simultaneously. The presented architecture consists of a layer of several LSTM units, one for each CAN ID, which their outputs are joined and fed into three fully connected layers as the bottleneck layers. Finally, the joint signals were reconstructed, and the reconstruction error was used as the anomaly score. This architecture allows the

network to consider all IDs' interdependencies. The authors tested the method on a real and a synthetic dataset, SynCAN. The authors suggested combining CANet with a rule-based IDS to improve its performance since the detection results were not in an acceptable range for some types of attacks [68].

In the multi-domain IDS proposed by Ashraf et al., the feature set was extracted from two datasets - the Car Hacking dataset and UNSW-NB15, representing the in-vehicle and inter-vehicle networks, respectively. Moreover, some extra statistical features were derived from these two datasets to improve the detection efficiency of the LSTM Autoencoder module. The method could detect attacks with more than 94% accuracy; however, since the UNSW-NB15 dataset is not VANET-specific, the inter-vehicle attack detection results cannot be generalized to the real environment [69].

Kukkala et al. introduced a novel self-attention mechanism into the LSTM autoencoder between the encoder and decoder parts and named it LATTE. The attention weights were computed by multiplication of the encoder outputs by linearly transformed hidden states, passing through a softmax function. Additionally, the encoder and decoder modules were composed of two LSTM layers. In the detector module, reconstruction errors of different signals in a message are aggregated, and the OCSVM algorithm is utilized to draw a hyperplane around normal case instances, separating them from anomalous errors. The efficiency of LATTE was tested considering various aggregation functions, and compared to static threshold detection, OCSVM had higher accuracy, precision, recall, and F-score. However, since the computational complexity of LATTE was relatively high, the authors suggested the distributed platform to overcome this challenge [72].

Other neural networks Other than the mentioned well-known neural network attack detection methods, some miscellaneous neural network structures are also offered. Among them is a GAN-based (Generative Adversarial Nets) IDS by Seo et al. It consists of a CNN-based generator and discriminator modules competing with each other to generate a fake data sample that is not detectable by the discriminator. The authors designed two discriminators, one for known attacks and one for unknown attacks [70]. Song and Kim suggested a noised pseudo-normal traffic generator based on LSTM prediction on the sequence of CAN IDs. They claim that if we have enough amount of normal data to train a model, the learned decision boundary is more reliable and can detect unknown traffics better. It was proved by applying the proposed generator on two anomaly detectors - DCNN and OCSVM [73].

The machine learning-based IDS for the in-vehicle domain, more specifically the CAN network, was initiated with statistical analytics and traditional machine learning

methods. Later, with the advances in deep learning, the research direction evolved towards the neural network-based anomaly detection systems for the CAN bus. Different NN structures, such as CNN, RNN, Autoencoder, and GAN are studied in the literature. Section 5 gives more information about future research directions, such as lightweight, distributed, collaborative, and blockchain-based IDS for the in-vehicle network. In addition, a summary of existing works on ML-based IDS for in-vehicle networks is provided in Table 1, sorted by the publication time.

3 Inter-vehicle network: security threats and IDS

3.1 Inter-vehicle communication

A vehicular ad hoc network (VANET) is a type of mobile ad hoc network, consisting of mobile nodes (i.e. individual vehicles) as well as stationary infrastructure nodes such as roadside units (RSUs) and certificate authorities [15, 74–76]. The vehicles can communicate wirelessly with each other (vehicle-to-vehicle or V2V communication), or with infrastructure nodes (vehicle-to-infrastructure or V2I communication) with the goal of reducing accidents, improving road safety, and reducing traffic congestion. Each vehicle in the network is equipped with an onboard unit (OBU) that enables communication with other vehicles (V2V) and infrastructure units (V2I).

A VANET architecture design consists of vehicles moving on roads at different speeds and RSU as shown in Fig. 2. In the figure, we can visualize the V2V and V2I communications sharing sensitive information between different components in VANET. Different communication protocols such as Dedicated Short-Range Communication (DSRC), Wireless Access in Vehicular Environments (WAVE), and cellular-V2X (CV2X) [77, 78] have been proposed for message dissemination in VANET. Vehicular communication is rapidly evolving to support traffic heterogeneity, for a variety of applications ranging from critical safety applications, such as forward collision warnings, lane assist systems, and traffic signal violation warnings to infotainment or comfort applications such as weather information systems, and gas/restaurant locations with price details. Safety applications rely on event-driven 'alert' messages (accident notifications) and periodic broadcasts of Basic Safety Messages (BSMs). BSMs contain critical information such as position, speed, and heading from participating vehicles. As the BSM data is used in real-time for high-speed applications, it is typically not encrypted to allow fast processing, which makes it vulnerable to attacks.

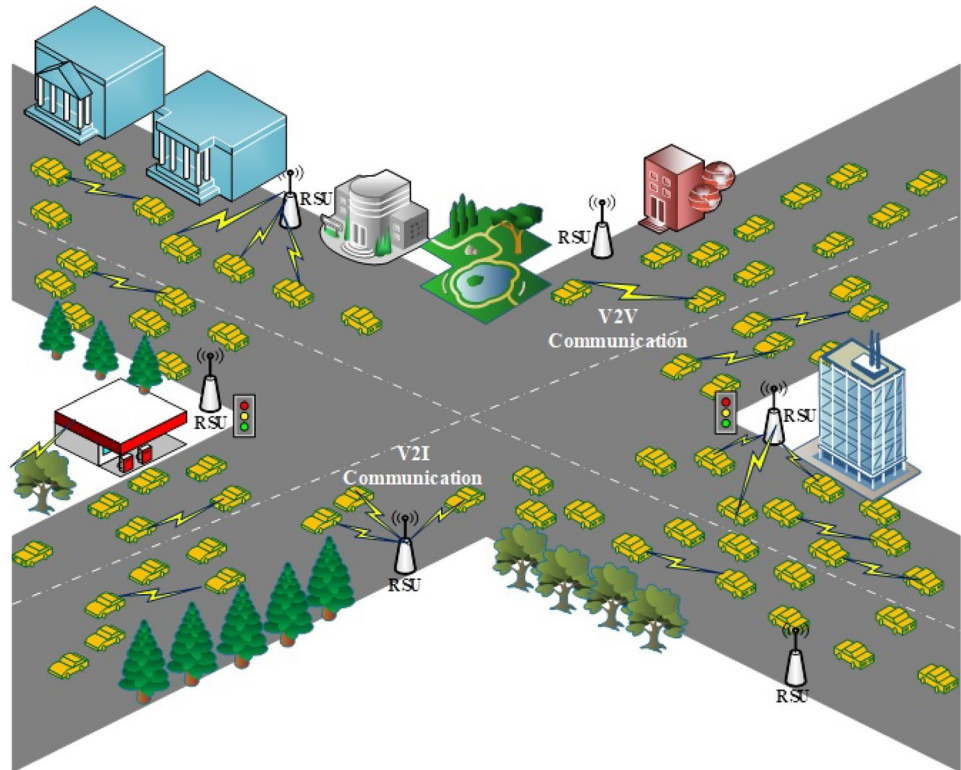
Table 1 A comparison of ML-based IDS approaches for CAN-BUS

Author/Paper	Year	Attack Type	ML/DL Model	Feature Set	Performance Metric
Theissler [38]	2014	Malfunctioning	OCSVM	Extracted signal	False negative per hour, True negative, True negative rate, Precision
Kang and Kang [47]	2016	Spoofing	DNN	Data Bit-symbol occurrences	False negative, False positive
Narayanan et al. [41]	2016	Malfunctioning	HMM	Extracted signal	N/A
Tian et al. [44]	2017	Spoofing	GBDT	Data, Entropy-based feature based on CAN ID and time	True positive, False positive
Levi et al. [42]	2018	Out of order scenario, USB firmware update attack, communication with unknown vendor, OTA malicious update, and Malicious application installation	HMM and regression	High abstraction of OS, network, and CAN interfaces	F-measure, AUC
Seo et al. [70]	2018	DoS, Fuzzy, Spoofing	GAN	CAN ID	Detection rate, Precision, Accuracy, AUC
Wang et al. [49]	2018	Replay, Spoofing	HTM	Data	Precision, Recall, AUC
Avatefipour et al. [39]	2019	DoS	OCSVM with Modified BAT algorithm (MBA) optimizer	CAN ID	Hit rate, Miss rate, Correct reject rate, False alarm rate, ROC
Zhang et al. [48]	2019	Replay, Spoofing	DNN with GDM/AG optimizer	Extracted signal	Time, ROC
Al-Saud et al. [40]	2019	DoS	OCSVM with Modified Social Spider Optimization (MSSO)	CAN ID, Frequency	Hit rate, Miss rate, Correct reject rate, False alarm rate
Casillo et al. [43]	2019	Malfunctioning	Bayesian network	Timestamp, Extracted signal	Precision, Recall, F-measure
Lokman et al. [64]	2019	DoS, Fuzzy, Spoofing	DNN Autoencoder	CAN ID, DLC, Data	Mean absolute error, Mean squared error
Xiao et al. [56]	2019	DoS, Fuzzy, Impersonation	RNN and attention mechanism	DLC, Data	Precision, Recall, F-measure, Accuracy, ROC
Zhu et al. [55]	2019	Spoofing, Replay, Flooding, Drop, Tamper	LSTM	Time interval, Data	Precision, Recall, F-measure, Accuracy, ROC, Time
Lin et al. [65]	2020	DoS, Fuzzy, Impersonation	Deep Denoising Autoencoder	CAN ID, Frequency	Error rate, Precision, Recall, F-measure, Hit rate, Miss rate, Correct reject rate, False alarm rate, Mean absolute error, Mean squared error
Jaoudi et al. [66]	2020	DoS, Fuzzy, Impersonation	CNN Autoencoder	CAN ID, Data	Precision, Recall, F-measure
Hanselmann et al. [68]	2020	Plateau, Continuous changing, Playback, Flooding, Suppress	LSTM	Extracted signal	AUC
Barletta et al. [71]	2020	DoS, Fuzzy, Spoofing	Kohonen Self-Organizing Map (SOM) and K-means clustering	CAN ID, DLC, Data, Time interval	Accuracy, Precision, Recall, F-measure, False negative rate
Song et al. [51]	2020	DoS, Fuzzy, Spoofing	DCNN	CAN ID	Precision, Recall, F-measure, False negative rate, Error rate
Hossain et al. [58]	2020	DoS, Fuzzy, Spoofing	CNN	CAN ID, DLC, Data	Accuracy, Recall, F-measure, False negative rate, False positive rate

Table 1 (continued)

Author/Paper	Year	Attack Type	ML/DL Model	Feature Set	Performance Metric
Hossain et al. [57]	2020	DoS, Fuzzy, Spoofing	LSTM	CAN ID, DLC, Data	AUC, Accuracy, Recall, F-measure, False negative rate, False positive rate
Longari et al. [67]	2020	Interleave, Discontinuity, Data field	LTSM Autoencoder	CAN ID, Data	Precision, Recall, F-measure, AUC
Ashraf et al. [69]	2020	DoS, Fuzzy, Spoofing	LSTM Autoencoder	Extracted features	Precision, Recall, F-measure, Accuracy, RoC
Yang et al. [46]	2021	DoS, Fuzzy, Spoofing	DT and K-means	Engineered features from CAN ID, DLC, and Data	Time, Memory usage, Accuracy, Detection rate, False positive rate, F-measure
Han et al. [45]	2021	Flooding, Fuzzy, Malfunction, Replay	DT, RandomForest, XGBoost	Mean, Variance, Q1, Q3, IQR, Skewness, and Kurtosis of event interval	Accuracy, Precision, Recall, F-measure, RoC, Time
Balaji and Ghaderi [59]	2021	DoS, Fuzzy, Spoofing	LSTM with Embedding	CAN ID, DLC, Data	RoC, True positive rate, False positive rate, F-measure
Kukkala et al. [72]	2021	Constant, Continuous, Replay, Dropping, DoS	LSTM Autoencoder and self-attention mechanism	Extracted signals	Accuracy, Recall, F-measure, False positive rate, RoC
Mehedi et al. [53]	2021	DoS, Fuzzy, Spoofing	CNN	CAN ID, DLC, Data	Precision, Recall, F-measure, Accuracy, AUC
Javed et al. [61]	2021	DoS, Fuzzy, Impersonation	CNN and attention-based GRU	Data	Precision, Recall, F-measure
Sun et al. [62]	2021	Replay, Drop, Spoofing, Fuzzy	CNN and attention-based Bi-LSTM	Extracted signals	Error rate, Precision, Recall, F-measure
Song and Kim [73]	2021	DoS, Fuzzy, Spoofing	LSTM	CAN ID	Precision, Recall, F-measure, Accuracy
Balaji et al. [60]	2022	DoS, Fuzzy, Spoofing	LSTM with Multi-Task Learning (MTL)	CAN ID, DLC, Data	F-measure, RoC, Memory usage, False positive rate, False negative rate, True positive rate, True negative rate
Thiruloga et al. [63]	2022	Plateau, Continuous changing, Playback, Suppress	TCNA (Temporal CNN with Attention)	Extracted signals	Detection Accuracy, ROC, F-measure, False positive rate, Mathews Correlation Coefficient

Fig. 2 Types of communications in VANET



3.2 Types of attacks

Inter-vehicle communication is susceptible to different types of threats, such as routing attacks, e.g. blackhole or wormhole attacks, data falsification attacks e.g. GPS spoofing, identity-based attacks e.g. impersonation and Sybil attacks, and denial of service attacks [79–81]. The need for sharing data quickly with many neighboring vehicles, infrastructure units, and other devices increases the attack surface for potential attacks. These attacks can compromise one or more of the security requirements for VANET communication, including authentication, confidentiality, data integrity, availability, etc. In this section, we will discuss some of the common attacks in each of these categories.

3.2.1 Attacks on authentication

Appropriate authentication protocols are needed for VANET communication to verify the security credentials of individual vehicles in a way that balances the privacy and accountability of the affected nodes. To limit the release of private information, vehicles typically communicate using authenticated pseudonyms [82], which are assigned along with public-private key pairs and certificates during registration. Some common attacks on authentication include Sybil attacks, node impersonation, and key replication.

According to Sheikh et al. [80], Sybil attacks, where the malicious attacker claims multiple identities to compromise the whole network, can be dangerous as an attacker has the potential to manipulate the behavior of other vehicles. Such attacks are difficult to detect and can lead to the interruption of services due to multiple messages using fake identities. In [83], Boudhir et al. discuss GPS spoofing attacks, where attackers can get unauthorized control of the VNS (Vehicle Navigation System) device and broadcast the wrong position of the vehicle. GPS spoofing can have serious consequences including traffic congestion and road accidents [74]. In node impersonation attacks, the attacker uses the valid ID of a registered vehicle and pretends to be a legitimate node, as described by Quyoom et al. [84]. For example in [85] attacker nodes send false messages to neighboring vehicles to disrupt traffic flow or claim to be an emergency vehicle so that other vehicles can stop or reduce the speed. Similarly, in key or certificate replication attacks, the attacker uses the key or certificates originally issued to other vehicles, in order to confuse the authorities and prevent them from identifying the actual attacker, as mentioned by Sheikh et al. in [80].

3.2.2 Attacks on availability

Any diminished capacity or unavailability of network resources can affect the proper functioning of safety

applications and pose potential harm to users in VANET [86]. DoS and routing attacks directly impact the availability of network resources and successful message delivery rates. DoS attack occurs when the attacker(s) interrupt the services by jamming the channel, overloading the network and/or dropping packets to leave the network fully or partially unavailable to legitimate users. It is one of the most common VANET attacks and can be carried out by both inside or outside attackers [87]. Kumar and Mann [88] identified two possible implementations of DoS attacks in VANET: i) the attacker repeatedly sends fake accident messages to the following vehicle and keeps it busy in the verification process instead of useful communication and ii) the attacker targets the RSU with the same strategy and leaves the communication unavailable for other vehicles. They observed that Packet Delivery Ratio (PDR), throughput, and end-to-end delay deteriorated significantly under the DoS attack. In Malebary et al. [89], the authors implement a jamming attack, using a hardware device to generate a fixed jamming power of 44dbm on channel 178 of the DSRC bands spectrum. Three attack scenarios are investigated, based on three mobility patterns of the attacker, via., stationary, random and target mobility. The attacker disturbs the radio signal's strength (power), which disrupts connections for legitimate vehicles and impacts the on-time reception of sensitive information such as accident, weather, and road condition notifications. Reactive jammers that target packets after they have been transmitted are hard to detect and can impact PDR and signal strength. In [90], Lachdehaf et al. implemented a blackhole attack, where the malicious node redirects the packets to a non-existing or non-identifiable node to prevent the safety messages from reaching the intended destination. AODV routing protocol along with route request message (RREQ) and route reply message (RREP) protocols was deployed to create a blackhole node in a simulation environment using NS3 and SUMO. They observed a negative impact on PDR, end-to-end delay, throughput, and routing overhead under the influence of blackhole attacks. In [91], the authors consider a new type of attack where the attacker does not alter the actual content of the message but injects a time offset into the original message to create a delay in reception. Using the Omnet++ simulator, they used a time-offset beacon send intervals of 0.1s, 0.15s, 0.20s, and 0.25s and observed an increase of up to 80% in delay. Timing attacks delay the delivery of time-critical safety messages, which can have severe negative repercussions [92].

3.2.3 Attacks on confidentiality and privacy

Cryptographic techniques, including the use of private and shared keys, are commonly employed to ensure the confidentiality of inter-vehicle communications [79]. Junejo et al. [93] presented a trust estimation method and decision

model to portray the eavesdropping attacks in VANET, where non-registered users attempt to access confidential data and resources. They considered four cases of “location closeness” (or trust zones), via., Roadside Trust Zone, Vehicle Trust Zone, and Vehicle Zone Coverage Area for both sender and receiver. The impact of the attack on the network, in the different trust zones, was then analyzed. Choudhari and Dorle [94] used Distribution Adaptive Distance with Channel Quality (DADCQ) approach to mount four different attacks including packet drop, packet manipulation, non-secure packet access, and eavesdropping. The results showed a decrease of around 3% in PDR after adding these attacks to the network. In [95], Ahmed et al. launched a Man-in-the-Middle attack using the VEINS simulator, with attackers being placed either randomly or together within a single fleet of vehicles. The number of malicious nodes was up to 50% of all nodes, and simulated attacks resulted in dropped packets, delay, and/or exchange of tampered messages among the registered vehicles.

Al-Shareeda et al. [96] consider attacks where a vehicle OBU uses the identity of another vehicle, computes the message signature, and broadcasts the message. They introduced single and batch verification schemes to distinguish messages sent using a fake identity; however, this requires additional computation and communication overhead during the attack phase. Vehicles often use temporary pseudonyms for V2V and V2I communication to protect the privacy and prevent eavesdroppers from learning their real identities [97]. Saini et al. [98] implemented eavesdropping attacks using PREXT [99] an extension of VEINS, based on the local passive adversary. They proposed intelligent adversary placement to evaluate the effectiveness of different privacy schemes in terms of the Global Tracking Success Rate (GTSR) for both urban and rural traffic scenarios.

In replay attacks [79], misbehaving vehicles broadcast a message that has been sent previously. Such attacks are typically able to foresee changes in time stamp and sequence number for each sent message [86] and can lead to inappropriate reactions in the vehicle systems such as Advanced Driving-Assisted System (ADAS). Kamel et al. [100] simulated a data replay attack using the Paris-Sacley traffic model with a low, medium, and high density of traffic, consisting of 12,542 vehicles and 8,475,371 V2X messages. In this work, an attacker chooses the target and replays its data to confuse the authorities. The authors also discussed different techniques to detect these attacks. In message fabrication, a malicious vehicle intentionally alters the contents of one or more messages. For example, an attacker can insert false position coordinates in Basic Safety Message (BSM) and Cooperative Awareness Message (CAM) [101]. In [101] Heijden et al. implemented a position falsification attack through simulations, using a Luxembourg traffic scenario. They generated a large dataset of BSMs containing

five different position falsification attacks, with attacker densities ranging from 10%–30% of total vehicles. Table 2 shows the comparison of various security attacks in terms of attacker model, compromised services, components, simulators used, and performance metrics to observe the impact of these attacks.

3.3 Attack detection approaches

Fast and accurate threat detection in VANET can be challenging due to high node mobility, unreliable link connectivity, and vulnerable, resource-constrained end nodes. Current detection can be broadly classified as data-centric (based on the message content) or node-centric (based on the message sender) or a hybrid of the two [104]. Attack detection can also be classified as collaborative or autonomous. Autonomous schemes analyze a stream of data from a single sender, while collaborative schemes compare messages from multiple nodes and try to detect anomalous transmissions. There exist different attack surfaces in VANET at local and global levels. OBU, Trusted Platform Module (TPM), GPS, and Sensors form an ad-hoc environment (local level) whereas Manufacturers, Certificate Authorities, Legal Authorities, and Service Providers are considered infrastructure environments (global level). RSU is the only component that works at both local and global levels. Security attack detection is crucial at both levels to ensure safety and efficiency in VANET.

The node-centric attack detection focuses on the behavior and reputation of the nodes sending the messages. Typical behavior analysis includes the format, frequency, and correctness of messages sent by a specific node. Hortelano et al. [105] proposed Watchdog as one of the primary detection techniques which utilize the behavior of vehicles in a multi-hop communication environment. Another example of behavior-based detection is the Flooding detection setup by Hamiah et al. [106] and Punal et al. [107]. In contrast, trust-based detection relies on trust scores assigned to nodes by surrounding nodes, including other vehicles and infrastructure units [57, 58]. Trust-based schemes simplify the revocation process for misbehaving nodes but can be more vulnerable to privacy threats. Data-centric attack detection methods analyze the contents of messages from a sender to ensure they are plausible and consistent with data received from other nodes. In [108] the authors use a statistical approach where each node calculates the average traffic flow in its neighborhood, based on messages from surrounding nodes and uses hypothesis testing to classify received messages.

The development of machine learning and deep learning models for VANET intrusion detection has been receiving increased research attention in recent years. These approaches use different ML algorithms, such as Support

Vector Machine (SVM), K-Nearest Neighbor (KNN), Random Forest (RF), etc., to analyze a particular set of packet features and classify attackers [57, 109, 110]. ML-based approaches can process a large number of records at a time, and often provide better detection rates compared to traditional techniques [102].

Deep Learning Based detection is a sub-class of ML-based detection, where deep learning algorithms, such as Convolutional Neural Networks (CNN), Multilayer Perceptron (MLP), Artificial Neural Networks (ANN), Deep Belief Networks (DBN) using Restricted Boltzmann Machine (RBM), the stacked sparse autoencoder (SSAE) with Softmax, etc., are used for detection of an attack. Deep learning has been shown to perform well in many domains, such as healthcare, chatbots, virtual assistants, robotics, etc. Deep learning algorithms analyze data with a logic structure similar to how a human brain would draw conclusions. It uses a layered structure of an Artificial Neural Network (ANN). The structure of ANN is inspired by the biological neural network of the human brain leading to a process of learning that's far more capable than that of standard machine learning algorithms. Deep learning algorithms typically require larger datasets for training and can produce better results. Deep learning also reduces the need for feature engineering tasks, which can take more time in machine learning-based approaches. Many deep learning approaches are well-suited for parallel processing, which can be advantageous for a large-scale environment [111].

3.4 Machine learning-based IDS for inter-vehicle communication

IDS for VANET aims to monitor network operations to detect different types of attacks. Standard cryptographic techniques including encryption/decryption and authentication can provide a level of protection against outsider attacks that do not have credentials to access the network but are ineffective against insider attacks. This can occur when a node with valid credentials becomes compromised. ML/DL-based approaches can be more effective in detecting such attacks and different techniques have been proposed in recent years. The majority of these works use supervised learning algorithms to detect known attacks; although some unsupervised learning and deep learning approaches have also been proposed in recent years.

Montenegro et al. [112] used K-nearest neighbor (KNN) algorithms, with a value of $K = 3$, to detect BSMs with false position data. They generated their own dataset containing four different false position attacks, i.e. fixed position, random position, fixed offset, and random offset, with a 50% probability that a BSM contains false data. The models achieved a detection rate of 91% for the fixed offset position attack, 86% for the fixed position, 91% for the random

Table 2 Security attacks in inter-vehicle communication for connected vehicles

Author/Paper	Year	Attack Type	Attacker Model	Compromised Services	Compromised Components of VANET	Simulators used	Performance Metric
Sumra et al. [92]	2011	Timing	Insider	Availability	Hardware, Software	N/A	Delay
Chhatwal and Sharma [85]	2015	Node Impersonation	Insider, Outsider	Integrity, Authentication	Infrastructure	N/A	N/A
Bhatti et al. [91]	2015	Timing	Insider	Availability	Hardware, Software	OMNET++	Delay,
Boudhir et al. [83]	2016	GPS spoofing	Outsider	Authentication	Sensors, Hardware, Software	three Arduino devices	N/A
Malebary et al. [89]	2016	Jamming	Insider	Availability	Sensors, OBU	NCTUNS 6.0	PDR, Signal Strength
Lachdhaf et al. [90]	2017	Black Hole	Insider, Malicious	Availability, Integrity, Confidentiality	Hardware, Software	NS2, SUMO, OSM	Delay, PDR, Throughput, Overhead
Ahmed and Elhadef [86]	2018	DoS	Insider, Outsider	Authentication, Availability	Wireless Interface, Hardware, Software, Infrastructure	N/A	Overhead, Load, Routing Stretch, Cumulative Distribution Frequency (CDF), Delay
Ahmad et al. [95]	2018	Man-in-the-Middle	Insider, Malicious	Confidentiality, Authentication, Integrity, Non-Repudiation, Privacy	Wireless Interface, Hardware, Software	OMNET, SUMO, VEINS	Delay, Content Delivery Ratio (CDR), Packet Loss Ratio (PLR)
van der Heijen et al. [101]	2018	Position Falsification	Malicious, Insider	Integrity, Privacy	Hardware, Software	OMNET, SUMO, VEINS	N/A
Kamel et al. [102]	2019	Sybil	Insider, Malicious	Authentication, Availability	Wireless Interface, Hardware, Software	OMNET, SUMO, VEINS	N/A
Singh et al. [103]	2019	Wormhole	Insider, Malicious	Availability, Integrity, Confidentiality	Hardware, Software	NS3, SUMO	Delay
Kumar and Mann [88]	2019	DoS	Malicious	Authentication, Availability	Wireless Interface, Hardware, Software, Infrastructure	NS2	Packet Loss, Throughput, PDR, Number of dead and alive nodes, Lifetime of Network
Choudhari and Dorle [94]	2019	Eavesdropping, DDoS	Insider	Authentication, Confidentiality, Privacy	Sensors, Hardware	N/A	PDR
Ilavendhan and Saruladha [87]	2020	DoS	Insider, Malicious	Authentication, Availability	Wireless Interface, Hardware, Software, Infrastructure	NS2, SUMO	Delay, Packet Loss
Saini et al. [98]	2020	Privacy	Insider	Confidentiality, Non-Repudiation	N/A	PREXT	GTSR
Kamel et al. [100]	2020	Data Replay	Insider, Malicious	Integrity, Authentication, Non-Repudiation	Hardware, Software	OMNET, SUMO, VEINS	N/A
Al-Shareeda et al. [96]	2020	Privacy	Insider	Confidentiality	N/A	OMNET, SUMO	Computation Overheads, Communication Overheads

Table 2 (continued)

Author/Paper	Year	Attack Type	Attacker Model	Compromised Services	Compromised Components of VANET	Simulators used	Performance Metric
Pu et al. [97]	2020	Privacy	Insider	Confidentiality	N/A	Blockchain	Computation Overheads, Communication Overheads
Junejo et al. [93]	2021	Eavesdropping	Insider, Malicious	Confidentiality, Privacy	Sensors	Mathematical Model	Location Closeness

position, and 84% for the random offset position attack. The validation dataset obtained an accuracy of 95%, recall as 100%, and ROC curve as 96%.

Grover et al. [109] used an ensemble supervised machine learning approach to detect a variety of attacks, including packet suppression, packet replay, packet detention, identity spoofing, and position forging. Attacks were generated using NCTUns-5.0 simulators [113] with attacker saturation of 25%, and focused on position, range, speed, and RSSI as the main features in the generated BSM dataset. Different algorithms, such as Naive base, Instance-Based Learner with parameter k (IBk), Decision Tree J-48, random forest, and AdaBoost were used to train and test data using WEKA [114], an online machine learning tool. Results indicate that random forest and J-48 performed the best in terms of true positive rate (TPR) and false negative rate (FNR), achieving an accuracy of 92% and 93% for binary class and multi-class setup respectively.

So et al. [115] used the VeReMi dataset [101] to detect position falsification attacks using KNN and SVM algorithms. The authors added a location plausibility check (feature 1) and movement plausibility check (feature 2) into the existing features in the VeReMi dataset and used 80% of the data for training and the remaining 20% for testing. In individual attack detection, SVM performed well for the detection of constant position and eventual stop attacks; whereas KNN had better detection rates for random position, constant offset, and random offset attacks. SVM also showed better performance with a combined dataset, where datasets for the different attack types were merged into a single file.

Sharma and Jaekel [116] proposed combining information from two consecutive BSMs from a sender to detect position falsification attacks using the VeReMi dataset. They extracted the features for all attack types and then removed non-contributing features and data points. They applied KNN, random forest, Naive Bayes, and decision tree algorithms on train and test datasets. KNN had the best overall performance for detecting single attacks and multi-class classification and achieved scores of over 98% in terms of precision, recall, and F1-score.

Singh et al. [103] applied SVM and KNN classification algorithms to detect wormhole attacks. The attack scenario and dataset were generated using SUMO [117] and NS3 [118] simulators and consisted of normal vehicles along with two malicious (attacker) vehicles. Each attacker used a tunnel to replay the packets to the other malicious node who then forwarded the same packet to the destination node. They used a modified version of the AODV protocol in NS3 and generated the dataset containing features such as SourceIP, destinationIP, BytesTransmitted, BytesReceived, BytesDropped, Delay, Jitter etc. The performance of the models were evaluated in terms of accuracy and false positive rate and it was demonstrated that both models could

achieve a high accuracy (99%) and a low false positive rate (14%).

Gyawali and Qian [110] used labeled data to detect two main categories of attacks i.e false alert attacks and position falsification attacks. They used realistic traffic scenarios using Open Street Map (OSM) [119] along with OMNET, SUMO, and VEINS [120] to generate the false alerts and true safety beacons. The dataset consisted of 11 different features including time, position, speed, etc. For position falsification attacks, they have used the publicly available VeReMi dataset. Various supervised machine learning algorithms, including logistic regression, KNN, decision tree, bagging, and random forest, were used for training and testing. Bagging classifiers achieved better accuracy (98%), compared to the other classifiers for both types of attacks.

Shams et al. [121] proposed Trust Aware SVM-based IDS (TSIDS), which utilizes trust value, SVM classification algorithm, and packet header monitoring to detect misbehaving vehicles. Trust value is a number assigned to each vehicle by a trusted authority. It can be increased or decreased based on the vehicle's behaviors in the network. Packet drop rate, packet transfer interval, and packet transfer delay are considered misbehavior parameters in this study. A private dataset generated, using SUMO, MOVE, and NS3 simulators, was used to train the SVM model, which achieved a 98% detection rate. The authors also reported an improved PDR from 77% to 87% and a reduction in end-to-end delay from 139 ms to 60 ms.

Sharma and Lui [122] introduced a data-centric approach for detecting location forgery by connected vehicles. They implemented different algorithms, such as SVM, KNN, Naive Bayes, random forest, ensemble-boosting, and ensemble-voting algorithms, and studied their performance with and without applying basic plausibility checks on location and mobility features in the VeReMi dataset [101]. Overall results show that precision increases by 5% and recall by 2% after including plausibility features. The classification errors for individual models ranged from 7.02% (voting) to 25.18% (Naive Bayes). The results highlight the advantages of ensemble learning models, which had lower classification errors, and are recommended for this type of attack detection.

Karagiannis and Argyriou [123] presented an RF jamming attack detection mechanism in VANET using a k-means clustering algorithm, an unsupervised ML approach for unlabelled data. In this experiment, they created three different scenarios of RF jamming attacks, including interference, smart attack, and constant attack. In an interference scenario, the attacker transmits the jamming signal from outside the network. In the smart scenario, the attacker follows the receiver-transmitter pair and transmits a jamming signal periodically from a secure location so that he can remain

undetected. In the constant scenario, the attacker follows the same strategy as in the smart scenario but does not bother to remain undetected. They introduced a new metric, Relative Speed Variation (RSV), derived from the relative speed of the jammer and target, as an extra feature to be used in the k-means clustering algorithm. The proposed model was able to successfully distinguish malicious and intentional jammers from unintentional jammers.

3.4.1 Deep learning based detection

Oucheikh et al. [124] proposed an anomaly detection model based on Convolution Neural Network (CNN) using the Wyoming Connected Vehicle Pilot Project dataset [125]. The dataset consists of BSM logs having 69 features such as longitude, latitude, position, speed, timestamp, and acceleration. They used Long Short-Term Memory (LSTM) ML algorithm for scaling, shuffling, and train/test split. After splitting the data, they used the train data to train an auto-encoder technique consisting of 6 layers to generate normal and attack features in the data. CNN is used for the classification of anomalies using contextual information (Temporal, Spatial, Spatio-Temporal) available in the provided BSM dataset. Results show that spatial-context has higher accuracy(95.7%), precision(94.8%), recall(93.9%), and specificity(95.4%) as compared to temporal or spatial context information.

Aneja et al. [126] proposed an IDS against the flooding attack using Artificial Neural Network (ANN) for classification and Genetic Algorithm (GA) for feature selection. Dataset traces (basic, IP, and AODV) were generated using NS2, SUMO, and MOVE to contain both normal (80%) and flooding attack instances (20%). The simulations revealed that PDR and throughput decreased after launching the attack. In the next stage, after data cleaning and preprocessing, they ran the ANN algorithm based on at least 3 layers and 22 neurons. The proposed approach obtained achieved high scores for accuracy (99%), precision (100%), specificity (100%), and F1-score (100%).

Kamel et al. [102] proposed a DL-based framework to detect Sybil attacks using Autencoder, Recurrent Neural Network(RNN), and LSTM. The dataset was generated using OMNeT, SUMO, and VEINS using the Luxembourg traffic scenario. The dataset has over 300 million BSM logs generated by over 100,000 vehicles and contains 30 features, including position, speed, acceleration, heading, time stamp, etc. The authors consider four types of Sybil attacks: Traffic Congestion Sybil, Data Replay Sybil, DOS Random Sybil, and DoS Disruptive Sybil. RNN, a deep neural network, successfully classified malicious BSM logs and achieved high values of accuracy(96%), precision(95%), recall(89%), and F1-score(92%).

Zeng et al. [111] presented DeepVCM, a deep learning-based IDS for connected vehicles. The dataset was generated using NS3 simulators and contained five different attack types; DoS, DDoS, Black hole, wormhole hole, and Sybil attacks. The simulation consisted of 300 normal and 20 misbehaving vehicles traveling over a 3000 m simulation area. One-dimensional CNN was used to extract information which is then passed as input to LSTM for the classification of attacks. The authors have used 2 convolution layers, 2 Maxpooling layers, and 2 Local Response Normalization Layers (LRM) for CNN and then, 3 layers of LSTM in their experiment. 80% of the data was used for training, 10% for validation and the remaining 10% for testing. Results show a better detection rate in terms of precision, recall, and F1-score for each of the four attack types compared to other IDS schemes mentioned in the literature [127].

Uprety et al. [128] incorporated a federated machine learning approach for Privacy Preserving Misbehavior Detection in the Internet of Vehicles (IoV). They used the VeReMi dataset [101] to train the ML model locally at each vehicle. The central authority (CA) holds the initial model using some ML model parameters and deploys the same model in randomly selected vehicles for local training. This approach can preserve the privacy of the data because vehicles do not need to share the data with any third-party entity. They performed the experiment using Artificial Neural Network (ANN) with a local batch size of 16 and a learning rate of 0.01. The proposed approach obtained better accuracy, precision, and recall when compared to the Stochastic Gradient Decent (SGD) approach. Table 3 gives an overview of different ML/DL-based approaches for misbehavior detection in VANET, including the types of attacks, specific ML/DL models and datasets used, and various metrics to evaluate the performance of the detection system.

4 Dataset and testbeds

Since ML algorithms learn from data, a comprehensive and valid set of data is needed for detecting intrusions in the connected vehicle's environment. The data must fully represent the real environment in terms of the dynamicity and diversity of normal scenarios and attack types. Researchers have chosen different approaches to obtain the data for training and evaluating the ML-based IDS, based on the type of network (in-vehicle or inter-vehicle), available computational power, the complexity of attack scenarios, and the scope of use cases. In this section, we will elaborate on these different available methods for testing the performance of IDS approaches in the CAV environment.

4.1 Datasets for VANET

Carrying out IDS performance evaluation through a dataset is a common and convenient approach in IDS research. While some studies make use of publicly available datasets, others might prefer to generate their own datasets by simulating the VANET environment. The simulation method is popular in the works done in VANET because of its flexibility to consider various scenarios. For connected vehicle scenarios, two types of simulators are used: a traffic simulator, for configuring the geography and mobility model of the environment, and a network simulator, for running network connections between elements and capturing messages from the network. NS3 and OMNET++ [130], as the network simulator, and SUMO [131], as the traffic simulator, are among the most frequently used simulators.

After selecting the traffic simulator, a test scenario should be chosen. This scenario aims to simulate the mobility and communication of vehicles in a VANET environment, based on some parameters: the simulated environment map (city, roads, highways, etc.), the topology for the mobility simulator (which roads a vehicle can ride in), the shape of buildings and parking areas (for a realistic simulation of wireless propagation model and communication channels), and mobility generation tools (for routing vehicles) [132]. There are test scenarios for different traffic simulators, namely the Sioux Falls scenario [133] for MATsim [134], TAPAS Cologne [135], VehiLux [136], and the LuST [132] for SUMO.

Despite the realistic and reliable results of the simulation method, using a dataset is more convenient because of its simplicity and lower computational power requirement. Some available datasets are used for IDS evaluation, including VANET IDS:

- NSL-KDD: is a revised and cleaned-up version of the KDD CUP 99 dataset, collected by the University of New Brunswick [137]. This dataset is widely used by researchers for testing network-based IDS approaches. Attack scenarios included in this dataset are DoS, Probe, U2R (user to root), and R2L (remote to use). He et al. [138], Kim et al. [139], and Gao et al. [140] relied on the NSL-KDD dataset for validation of their machine learning-based IDS for VANET.
- UNSW-NB15: is a network flow-based dataset presented by Moustafa et al. in the Cyber Range Lab of the Australian Centre for Cyber Security (ACCS) [141]. This dataset includes 49 features and nine attack types, including Fuzzers (injecting a massive number of random messages), Analysis (port scan, spam emails, etc.), Backdoors (bypassing authentication), DoS, Exploits, Generic, Reconnaissance (gathering information as a

Table 3 Comparison of ML/DL Based IDS Approaches in Connected Vehicles

Author/Paper	Year	Attack Type	ML/DL Model	Dataset Name	Dataset Available?	Performance Measures
Grover et al. [109]	2011	Packet forging, Identity spoofing	Naive Bayes, IBk, Decision Tree, J-48, Random Forest, AdaBoost	Simulated	No	Accuracy, True positive rate, False negative rate
Karagiannis and Argyriou [123]	2018	RF Jamming	k-means Clustering	Simulated	No	Relative Speed Variation (RSV), Clusters
Aneja et al. [126]	2018	Flooding Attack	ANN, GA	Simulated	No	Accuracy, Precision, recall, F1-Score
Shams et al. [121]	2018	Packet Drop, Packet Transfer Delay, Packet Ransfer Interval	SVM	N/A	N/A	PDR, Delay, Accuracy
So et al. [115]	2018	Position falsification	kNN, SVM	VeReMi	Yes	CCR, Precision, Recall
Singh et al. [103]	2019	Wormhole Attack	SVM, kNN	Simulated	No	Accuracy, True Positive Rate, False Positive Rate
Gyawali and Qian [110]	2019	Position Falsification	Logistic Regression, kNN, Decision Tree, Bagging, Random Forest	VeReMi	Yes	
Kamel et al. [102]	2019	Data Replay, Sybil, DoS	Autoencoder, LSTM, RNN	VeReMi Extension	No	Accuracy, Precision, recall, F1-Score
Zeng et al. [111]	2019	DoS, DDos, Black hole, Wormhole, Sybil	LSTM, CNN	Simulated	No	Accuracy, Precision, recall, F1-Score
Montenegro et al. [112]	2020	Position Falsification	kNN with k = 3	Simulated	No	Accuracy, Recall, ROC
Sharma and Liu [122]	2020	Position Forgery	SVM, kNN, Naive Bayes, Random Forest, Ensemble-Boosting, Ensemble-Voting	VeReMi	Yes	Precision, Recall
Oucheikh et al. [124]	2020	Anomaly Detection	CNN, LSTM	Wyoming	Yes	Accuracy, Precision, recall, F1-Score
Sharma and Jaekel [116]	2021	Location Spoofing	kNN, Random Forest, Naive Bayes, Decision Tree	VeReMi	Yes	Precision, recall, F1-Score
Upreti et al. [128]	2021	Position Falsification	ANN, SGD	VeReMi	Yes	Accuracy, Precision, recall
Jaton [129]	2021	DDoS	Multi Layers Perceptron (MLP)	Simulated	No	Accuracy, F1-Score

probe), Shellcode and Worms. UNSW-NB15 is used in [140] and [69] as the evaluation method of their proposed IDS for VANET.

- Kyoto dataset: is generated by Kyoto University using a honeypot platform for evaluating IDS performance [142]. This dataset consists of 14 conventional features, which are extracted based on KDD-CUP, as well as 10 additional features. Alheeti et al. used this dataset to validate their NN-based IDS for VANET [143].
- ISCX 2012 IDS: is presented by the Information Security Centre of Excellence (ISCX) of the University of New Brunswick. It contains five categories of traffic: normal, DDos, Brute Force SSH, HttpDoS, and Infiltrating Transfer [144].

The mentioned datasets are publicly available, frequently used for IDS evaluations, and include a wide range of attacks. However, their traffic data is not VANET-specific and doesn't fit the VANET environment.

To overcome the shortage in the VANET-specific datasets, the VeReMi dataset was presented by van der Heijden et al. [101]. VeReMi has successfully simulated a comprehensive and VANET-specific dataset by using simulation. VEINS [120], an open-source framework for running inter-vehicle simulations, based on SUMO (as the traffic simulator) and OMNET++ (as the network simulator), is utilized for generating VeReMi. Also, the LuST is utilized as the test scenario for the SUMO. The simulated attacks in VeReMi include constant attack, constant offset attack, random

attack, random offset attack, and eventual stop attack. However, this dataset does not include the DoS attack.

Later, VeReMi was extended to include more attack types, namely DoS, DoS random, data replay, disruptive, and traffic congestion Sybil attacks in V2V communications. It also became more realistic by taking into account sensor errors, sensor malfunctioning, and delayed messages [100].

Some other studies generated datasets that are not publicly available, but they can be regenerated based on the specification in their work. Generally, the approach goes through three stages: first, running a vehicle mobility simulation, second, capturing the normal network traffic by a network simulator, and finally, injecting malicious data into the dataset.

Belenko et al. proposed a dataset generation approach using an NS3 simulator [145]. This work is considered to be a baseline for other works, using the simulation for assessing the performance of their proposed IDS. In this approach, the first step is to define input parameters for NS3 VANET simulation, including location, the scenario of threats, the number of VANET hosts, and the number of intruders in the simulated environment. Then the simulation is started based on the defined parameters, and three outputs are generated, which can be used as datasets. These outputs are routing tables of each VANET host, captured packets, and network statistics. This approach can generate DoS, DDoS, wormhole, gray hole, black hole, and Sybil attacks.

Authors in [146] presented a dataset for VANET IDS, including both normal and attack data. This dataset was generated by NS3 as the network simulator and SUMO as the traffic simulator, while a Java-based platform, called VSimRTI [147], is used to couple these two simulators. The attacks implemented in this dataset are DoS and Fabrication attacks for speed, acceleration, and fake heading. Their proposed dataset is available for researchers at [148].

A dataset for detecting DDoS attacks in VANET is generated by [149]. Their generated dataset consists of 29 features captured from a simulation run by SUMO and OMNET++. The simulation process and parameters are documented clearly and can be regenerated for comparison purposes.

Zeng et al. has used both simulation and dataset to validate their deep learning-based IDS in VANET [111]. The dataset used in this study is ISCX 2012 IDS dataset. For the simulation part, they generated their dataset by NS3 simulator, following the dataset generation process in [145]. The only difference is that they only captured raw packets, not the features.

In the DT and deep belief machine learning IDS proposed by [150], also both NS3 simulation and NSL-KDD dataset are used for evaluation. They have simulated normal traffic by NS3 on a VANET environment consisting of 40 vehicles running for 600 s. Then NSL-KDD dataset is used to generate DDoS, DoS, Probe, U2R (user to root), and R2L (remote to use) attack scenarios.

Authors in [124] proposed an unsupervised clustering IDS for detecting anomalies in VANET using BSM messages. They utilized BSM from Wyoming Connected Vehicle (CV) Pilot project [151]. This dataset is collected from V2V communications and contains 69 features, like longitude, latitude, altitude (position), speed, timestamp, and acceleration.

A comparison of various datasets utilized for evaluating VANET IDS is shown in Table 4. Datasets are analyzed through the network and traffic simulators they used, dataset features, implemented attack types, VANET-specificity, and availability to researchers.

4.2 Datasets for CAN-BUS

For evaluating IDS proposed for CAN-BUS security, the most common approach is using datasets. The main challenge of studying CAN-BUS security is the lack of publicly available datasets. However, HCR Lab has introduced two publicly available datasets. One is the CAN IDS dataset [34]. It is used in many recent studies like [39, 46, 65, 69, 71, 152], and contains DoS, Fuzzy, and Impersonation attack scenarios. The data is collected through the Raspberry Pi3 and Arduino devices connected to the OBD-II port of a real vehicle and logging CAN messages. Fuzzy and DoS attacks are implemented using Raspberry Pi3 with PiCAN2, and Arduino with CAN-Shield is used to generate impersonation attacks. Table 5 illustrates an example of the DoS attack implemented in the OTIDS dataset.

The other public dataset for CAN-BUS from HCR Lab is the Car Hacking dataset, proposed by [51]. The data is collected by connecting two Raspberry Pi devices to the OBD-II port of a real car, one for logging the network traffic and one for injecting fabricated messages to the CAN-BUS. The dataset consists of four datasets: normal, DoS, fuzzy, and spoofing RPM gauge and gear. It is a publicly available dataset that is widely used by authors working on CAN-BUS security. Each message in this dataset consists of a timestamp, CAN ID, DLC, DATA[0:7], and a flag indicating normal (R) or attack data (T). An example of a DoS attack scenario in the Car Hacking dataset is shown in Table 6.

Automotive CAN bus intrusion dataset v2 is another available dataset for CAN intrusion detection offered by TU Eindhoven [153]. CAN messages are generated by two actual vehicles, Opel Astra and Renault Clio, and captured from the CAN-to-USB interface (CANtact) connected to their OBD-II ports. A prototype environment is also simulated using two Arduino boards (one programmed as legitimate and the other as compromised ECU) with CAN bus shields and a joystick acting as the throttle. The attacks are injected manually into the regular traffic. These attacks are fuzzy, replay, suspension, DoS, speedometer spoofing, and diagnostic [154] attacks.

Table 4 Datasets in VANET

Dataset Name	Reference	Network Simulator	Traffic Simulator	Features	Attack Types	VANET Specific	Available
NSL-KDD	Tavallae et al. [137]	N/A	N/A	41 features, including Basic, Content, Timing, and Host features	DoS, Probe, U2R, and R2L	No	Yes
UNSW-NB15	Moustafa and Slay [141]	N/A	N/A	49 features, including Basic, Flow, Timing, Content, Connection, and Additional features	Fuzzing, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worm	No	Yes
Kyoto	Song et al. [142]	N/A	N/A	14 features from NSL-KDD and 10 additional features		No	Yes
ISCX 2012 IDS	Shiravi et al. [144]	N/A	N/A	Full packet captured	DDoS, Brute Force SSH, HtpDoS, and Infiltrating Transfer	No	Yes
VeReMi	van der Heijden et al. [101]	OMNET++	SUMO + LuST	Time stamp, transmission time, sender, message ID, position, a speed, the RSSI, position noise and a speed noise	constant attack, constant offset attack, random attack, random offset attack, and eventual stop attack	Yes	Yes
VeReMi Extension	Kamel et al. [100]	OMNET++	SUMO + LuST	Type, transmission time, sender, message ID, position, speed, acceleration, heading, position noise, speed noise, acceleration noise, heading noise	Constant attack, Constant offset attack, Random attack, Random offset attack, Eventual stop, DoS, DoS random, data replay, disruptive, and traffic congestion Sybil attacks	Yes	No
N/A	Belenko et al. [145]	NS3	N/A	routing tables, captured packets, and network statistics	DoS, DDoS, wormhole, gray hole, black hole, and Sybil attacks	Yes	No
N/A	Gonçalves et al. [146]	NS3	SUMO	Timestamp and Vehicle ID for send messages, Timestamp, Sender and receiver ID, Time difference, heading, speed, acceleration, position, and differences of heading, speed, acceleration, position	DoS and Fabrication attacks for speed, acceleration, and heading	Yes	Yes
N/A	Alhaidari and Alrehan [149]	OMNET++	SUMO	29 features, including packet information and network statistics	DDoS	Yes	No

Table 5 DoS attack in CAN IDS dataset. CAN message with the highest priority (ID=0X00) are injected into normal data to prevent the processing of other legitimate messages from ECUs

Timestamp	ID	Control	DLC	Data								
0.000000	0000	000	8	00	00	00	00	00	00	00	00	00
0.000271	0080	000	8	00	17	dc	09	16	11	16	bb	
0.000495	0000	000	8	00	00	00	00	00	00	00	00	00
0.000736	0081	000	8	40	84	87	00	00	00	00	00	6b
0.000983	0000	000	8	00	00	00	00	00	00	00	00	00
0.001239	0165	000	8	00	08	80	00	00	00	00	08	80
0.001484	0000	000	8	00	00	00	00	00	00	00	00	00

SynCAN is a synthetic CAN messages dataset generated by Hanselmann et al. to evaluate their CANet NN-based IDS for CAN-BUS [68]. Unlike other datasets, this one consists of a diverse range of attack types and translated payload signals. The attacks include plateauing (constant value), continuous changing (slight drift in data), playback, flooding, and suppressing (preventing ECUs from transmitting data). As another characteristic of this dataset, the raw binary payload of CAN messages is decoded into their original signals. For each ID, a so-called CAN matrix specifies which bits encode which signal in the payload. CAN matrix is an intellectual property of most car manufacturers, which makes it impossible for others to decode CAN payloads. In spite of SynCAN providing the decoded signals, the designed ML-based IDS trained on this dataset may not be applicable to other manufacturers. SynCAN is available online at [155].

The Oak Ridge National Laboratory (ORNL) has published the ROAD (Real ORNL Automotive Dynamometer) dataset for CAN intrusion detection. ROAD is a real car dataset that provides both the raw binary payload and extracted signals [156]. The real attacks in this dataset are categorized into fabricated messages attacks, including fuzzy and targeted ID spoofing attacks, in which a portion of the payload is manipulated and injected, and non-fabricated messages attacks, which are more advanced ones. Also, a masquerade attack is simulated by deleting some normal messages. All attacks are verified physically to observe their effects. This is the most recent and comprehensive available dataset, containing real, verified, and more complicated attacks compared to the others. However, a DoS attack is not implemented in it.

Some authors have gathered their own datasets in addition to these publicly available ones. Tian et al. used a real car

dataset, called Alsvin CHANA, a dataset of 750,000 CAN-BUS messages, to evaluate their decision-tree IDS [44]. They gathered normal data from a car at a slow speed and changed some of the values randomly to a number between 0 and 255 for generating anomalies in the data. However, their dataset is not publicly available because of information security and privacy issues. The other limitation of this dataset is that it only contains payload modification attacks, but not DoS attacks.

ZOE dataset proposed by Rieke et al. is another dataset containing CAN-BUS messages [35]. It is normal state data, logged from the OBD port of the Renault Zoe electric car driving in an urban environment for 10 min. They injected falsified steering, brake, and speed information into the data as an injection attack, and also a set of messages increasing and decreasing the vehicle speed, as they seem to be fast acceleration or emergency brake. This dataset is also not available online. Berger et al. used the ZOE dataset, along with the Car Hacking dataset, for comparing the efficiency of different ML approaches [157].

Al-Saud et al. proposed a modified social spider optimization IDS and used their simulated environment for carrying out the evaluation [40]. In this simulation, CAN messages were captured from a real electric car. The car was turned on, kept idle, and after a few seconds, driven for 8 min. While driving, the brake was hit, then stopped, and backed up. Finally, after a few seconds of staying in parking mode, the vehicle is turned off. The attack messages are also injected with a specific ID. Like the other mentioned studies, the dataset in this paper is not available online.

Avatefipour et al. also used their simulation and two public CAN-BUS datasets, CAN IDS and Crash Reconstruction Research Consortium of the University of Tulsa [39].

Table 6 DoS attack in Car Hacking dataset. DoS attack messages, with ID=0X00, are labeled with T

Timestamp	ID	DLC	Data									Label
1478198794.437161	04f0	8	00	00	00	80	00	69	d1	13		R
1478198794.442596	0316	8	05	21	40	0a	21	1e	00	6f		R
1478198794.448032	0002	8	00	00	00	00	00	03	0a	0c		R
1478198794.453865	0000	8	00	00	00	00	00	00	00	00		T
1478198794.459302	0000	8	00	00	00	00	00	00	00	00		T
1478198794.464913	0000	8	00	00	00	00	00	00	00	00		T

The second dataset consists of CAN messages of the Dodge RAM Pickup car, driving ordinary in a residential area.

Hussain et al. also generated their dataset for CAN-BUS, called NAIST dataset [57]. NAIST is generated by collecting CAN messages from a Toyota hybrid car, working ordinarily for 120 s. Three types of anomalies, namely DoS, spoofing, and fuzzing, are injected into the normal data.

In [42] an event-based dataset is proposed by Levi et al., adding a layer of abstraction to the data collected by vehicles. SUMO is used for the simulation of vehicle mobility and they used a data collector for each car in the simulation. Instead of raw CAN messages, a sequence of occurred events in each vehicle called a story, is used for detecting anomalous behavior. These events include physical events like login, logout, door open, door close, and application-wide events like file access, network usage, app update, etc. Due to the event-based characteristic of data, the attack scenarios they implemented are defined as a substantial variation in the usual driving stories. Implemented attack scenarios are: out of order scenario, when an event appears in the wrong order, USB firmware update attack, when a USB abnormally is ejected and inserted in the middle of an expected firmware update story, communication with an unknown vendor, OTA malicious update, and malicious application installation. The novelty of this dataset is its event-based higher-level interpretation and detection of attacks; however, DoS attacks cannot be detected using this dataset, and also it is not publicly available to researchers.

Table 7 represents an overview of datasets used in detecting anomalous behavior in CAN-BUS, as well as their data acquisition approach, features, and attack types.

4.3 Testbed

A testbed is a combination of hardware and software modules designed to emulate a testing environment for a new product, study the interaction of different components, and thus have an insight into the efficacy and efficiency of the proposed system. The testbed is the evaluation platform for some research areas, especially security and safety assessments. CAV testbeds categorize into hardware-based, simulation-based, and hybrid testbeds. Hardware-based testbeds utilize the physical hardware modules, for instance, real vehicles, sensors, roads, base stations, etc., while in the simulation-based testbeds, the product is tested by software components that simulate the driving environment and conditions, reducing the risk and cost of experiments. A hybrid testbed is a mixture of hardware and simulation-based methods and benefits the reliability and cost-effectiveness of both [158].

Some laboratories have offered hardware-based testbeds to test connected vehicles for industry and research purposes. California CV Testbed is the first public testbed founded in 2005 at UC Berkeley. The testbed is located in

the heart of Silicon Valley in Palo Alto and was updated in 2018 to be aligned with the latest CV technologies and standards. 4 G/LTE technology is used as the backhaul communication between OBUs in vehicles and RSUs on the roads; however, connected vehicles require a 5 G communication infrastructure [159].

Millbrook testbed is also a CAV test laboratory located in the USA (Michigan and California), the UK, and Northern Finland. The testbed provides facilities for testing connected vehicles using cellular and 802.11p connections on the 4 G LTE and 5 G testbed, provided by Zenzic (the Centre for Connected and Autonomous Vehicles (CCAV)). The data gathered by the test environment is available and can be viewed in real-time worldwide [160].

While testing connected vehicles in a real-world driving car scenario seems more realistic and reliable, sometimes emulating a specific driving scenario is not feasible or safe. For example, in a pedestrian collision-avoidance scenario, failure of the test might result in injuries and damages. The Simulation-based testbed enables the researchers to validate their CV solutions in a secure environment.

OCTANE (Open Car Testbed and Network Experiments) is a simulation-based testbed platform that enables researchers to set up a test intra-vehicle environment [161]. It can be used for research and teaching purposes, supporting CAN, LIN, and FlexRay. It is made up of a software package and hardware. The GUI makes it easy to operate and translate user inputs into CAN messages. The messages on BUS are logged into XML format for further analysis and evaluation. The hardware setup can be designed based on the type of research, network type, automobile type, and budget. OCTANE testbed platform is used by Kang et al. to generate a normal dataset for evaluating their DNN IDS approach [47].

CARLA is an open-source simulator software for autonomous driving [162]. It can support different sensors like LIDARs, multiple cameras, depth sensors, and GPS. It has a graphical interface, as well as well-defined APIs, to control the simulation environment situations in terms of weather, sensors, obstacle behavior model, etc. Cassilo et al. used This simulation method to evaluate their Bayesian Network-based IDS for CAN-BUS security [43].

The other option for configuring a testbed is the hardware-in-the-loop (HIL) simulation in the lab. In this kind of simulation, a hardware sensor or even a whole vehicle is driven in a simulated driving environment. HIL scenario can be considered as a hybrid testbed comprising a hardware module (car and sensors) in a simulated driving environment.

Spirent has presented the GSS7000 [163] solution along with the SimHIL [164] to simulate a Hardware-in-the-loop scenario for connected vehicles. GSS7000 is a hardware module for testing GNSS and motion tracking systems. SimHIL is a powerful API for integrating GSS7000 with

Table 7 Datasets in CAN-BUS

Authors	IDS	Data Acquisition Approach		Dataset Name	Dataset Features	Attack Types	Available
		Simulation	Real Car				
Lee et al. [34]	OTIDS		x	CAN IDS	timestamp, CAN ID, DLC, DATA[0~7]	DoS, Fuzzy, Impersonation	Yes
Song et al. [51]	DCNN		x	Car Hacking	timestamp, CAN ID, DLC, DATA[0~7]	DoS, Fuzzy, Spoofing (RPM and Gear)	Yes
Tian et al. [44]	DT IDS		x	Alsvin CHANA	timestamp, CAN ID, DLC, DATA[0~7]	Spoofing	No
Rieke et al. [35]	Petri Net		x	ZOE	timestamp, CAN ID, DLC, DATA[0~7]	Spoofing	No
Al-Saud et al. [40]	MSSO		x	N/A	CAN ID, Message frequency	DoS	No
Avatefipour et al. [39]	MBA-OCSVM		x	CAN IDS, Crash Reconstruction Research Consortium	CAN ID, Message frequency	DoS	No
Hanselmann et al. [68]	CANet NN-based IDS		x	SynCAN	timestamp, CAN ID, DATA[0~7]	Plateau attack, Continuous changing attack, Playback attack, Flooding attack, and Suppress attack	Yes
Hossain et al. [57]	LSTM IDS		x	NAIST	CAN ID, DLC, DATA[0~7]	DoS, Spoofing, and Fuzzy	No
Levi et al. [42]	HMM IDS	x		N/A	Events	Out of order scenario, USB firmware update attack, Communication with Unknown vendor, OTA malicious update, and Malicious application installation	No
Kang and Kang [47]	DNN IDS	x		N/A	CAN ID, DLC, DATA[0~7]	Spoofing	No

the dynamic HIL environment. Together, they can play a part in simulating driving car mobility on a simulated road, facing obstacles, pedestrians, and traffic lights, and collecting data from sensors, CAN-BUS, and V2X communications.

Another HIL solution is provided by Danlaw MxDriver Mobility Simulator [165]. In this product, various parameters can be configured for implementing different test scenarios, including routes and maps, infrastructure transmissions, types of vehicle BUS, and other actors like people and bicycles.

5 Future research directions

5.1 Lightweight IDS for in-vehicle network

Existing approaches for CAN IDS currently primarily use fingerprint-based techniques, parameter monitoring, and information-theoretic approaches. These techniques' detection performance is limited and cannot be effective against new/unknown attack scenarios. In addition, for in-vehicle scenarios, real-time detection is necessary and constrained computing resources should be considered. Model

compression and acceleration approaches will be applied to the deep learning models to reduce the storage and computational cost for some real-time detection. But with evolving Functional Safety Standards based on ISO 26262 and ISO/SAE 21434, the systems are designed to eliminate any possible threats, means to react in the event of cyber threats, and make ECUs robust for deploying IDS based on Lightweight ML models. The proposed IDS models will be lightweight, computationally efficient, and specifically designed for systems with limited computing and storage resources

5.2 Distributed learning model for IDS

Typically, machine learning-based models for IDS are trained at either a single-user server or a centralized server. However, those solutions require the whole data to be sent to a central server for training, which can incur potential security and privacy risks during transmission and data processing at the central server. Moreover, it is computationally expensive to process a huge volume of data on a single server. To address those issues and limitations, one of the promising approaches in the machine learning-based approach is federated learning. In federated learning, machine learning models will be trained in a distributed manner, which keeps the data locally on the edge devices such as vehicles or RSUs, and only sends the learned machine learning model weights to the central server. Decentralized federated learning approaches can be investigated to enable anomaly detection in vehicular networks. Specifically, federated learning enables local training for anomaly detection in vehicular networks without the necessity of sending raw data to the centralized server.

5.3 Cooperative intrusion detection for CAVs

Attacks can occur in different parts of the vehicular networks simultaneously (e.g., distributed DoS attacks). In addition, it is more difficult to detect coordinated attacks using isolated IDS, because those individual IDS typically only monitor a small part of the network. Moreover, although a particular IDS can be very efficient to detect certain attacks, the detection performance can be much worse for different attacks, compared with other IDS. To improve the detection performance and better protect the vehicular networks, the collaborative intrusion detection paradigm has the potential to detect intrusions that occur across the network, by correlating results from different IDSs. Approaches such as ensemble-based collaborative IDS can help accelerate attack identification, reduce false alarm rates, and increase detection rate. The ensemble-based techniques can be adapted to combine multiple models in order to classify new instances using a combination rule, such as the majority voting rule and average rule.

5.4 Blockchain based collective learning for CAVs

A blockchain for the Internet of Vehicles is a distributed database, which means that the database's storage devices are not connected to a central processor, but rather are composed of encrypted interconnected blocks with consensus algorithms and smart contracts to ensure the highest level of data protection for the vehicle's connected environment. Rather than relying on a central authority, these blocks are formed up of transactions that are independently reviewed and confirmed. Immutability, transparency, disintermediation enables blockchain to be efficiently incorporated with other machine learning and federated learning techniques in transfer, store, provide validation between two nodes. The blockchain's capacity to track and verify data is another significant characteristic. There are now three options for implementing blockchain on a network: public, private, and consortium. The most typical implementation of cryptocurrencies is the public blockchain, which allows every node to engage in some activity of the management process. The nodes and rules of a private implementation are under the jurisdiction of a single organization. Finally, there is the consortium blockchain, where a collection of firms, generally with a same goal, administer and govern the nodes and their regulations on behalf of the consortium. According to a survey of academics, scholars prefer consortium blockchains to other forms of blockchain. Because public blockchains are completely decentralized, each node must execute every transaction. This means that the scalability of public blockchains is often limited. There is also the issue of storage on public blockchains, as each node must keep every block of the immutable blockchain. Unlike public blockchains, private ones don't face many of the same problems. There is a central authority that decides which nodes may join the blockchain, which nodes can be miners, which nodes must contain all transactions, etc. However, the entire blockchain is controlled by a single business, making it more centralized. In a consortium blockchain, the blockchain is only partially private since it is owned by many firms and a consensus algorithm. Thus, the blockchain-based approach proves to be more promising for the IoV environment and provides undivided intrusion detection services.

6 Conclusion

This article provided a comprehensive survey on machine learning-based IDS for securing CAV. Specifically, we discussed the security threats in CAV and presented the recent advances for machine learning-based IDS to address the security issues, from the perspectives of intra-vehicle and inter-vehicle networks, respectively. We also discussed available datasets for IDS in CAV and summarized the

current testbeds. Open research issues were outlined for further investigation. This survey expects to shed light on IDS design in CAV and attract more research efforts on machine learning-based solutions for safeguarding CAV.

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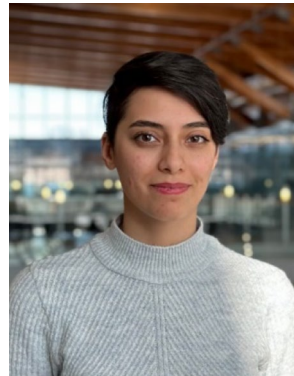
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